

Diagnostic Inflation Expectations in China: Salience, Overreaction, and the Limits of Rational Updating*

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Abstract

Do Chinese households overreact to salient inflation signals? I construct a quarterly panel from the People's Bank of China depositor survey (2011Q1–2025Q3) and complement it with seven waves of the China Family Panel Studies to test the diagnostic-expectations hypothesis at both aggregate and household levels. A salience-IV design, where media congestion instruments inflation salience, traces a testable chain: salience shifts expectation revisions, and revisions predict subsequent forecast errors with a negative sign ($\hat{\beta} = -2.24$, $p = 0.055$, $N = 32$), consistent with diagnostic overreaction. Local projections show that revision shocks move forecast errors within one quarter and then attenuate. A mechanism-competition design reveals that both diagnostic and information-rigidity channels carry explanatory power, with the diagnostic margin activating more strongly during high-uncertainty episodes. Heterogeneity analysis using CFPS microdata shows that forecast errors conditional on revisions are larger among less-educated, lower-income, and rural households, consistent with attention-allocation models. A Bayesian state-space model tracks time variation in the diagnostic coefficient, which intensifies during geopolitical crises and pandemic onset. Translating these estimates into a real-time forecast-adjustment rule improves out-of-sample MAE by 17% and

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interval coverage from 42% to 67% relative to naive survey forecasts. The evidence supports diagnostic overreaction as one operative channel in Chinese household expectation formation, while external validity remains bounded by national aggregation and short-horizon survey design.

JEL Codes: D84, E31, E71, G41

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1 Introduction

Household inflation expectations shape monetary policy transmission, yet the cognitive process behind those expectations remains contested. This paper tests whether Chinese households overreact to salient inflation signals in the manner predicted by diagnostic expectations (Bordalo et al.), and whether the resulting forecast errors are economically meaningful and predictable.

I find that they do. Using quarterly data from the People’s Bank of China Urban Depositor Survey (2011Q1–2025Q3) and seven waves of the China Family Panel Studies (CFPS, 2010–2022), I document a negative revision–error coefficient at both the aggregate and household levels. At the aggregate level, quarters in which households sharply revise inflation expectations upward are followed by negative forecast errors as realized inflation undershoots the revised belief ($\hat{\beta}_1 = -2.24$, Newey–West $p = 0.055$, $N = 32$). At the household level, the CFPS panel confirms this pattern with substantially greater statistical power: the revision coefficient is approximately -0.55 ($p < 0.001$, $N \approx 13,700$) with province and wave fixed effects. Both estimates carry the sign predicted by diagnostic overreaction and are inconsistent with the rational-expectations null of zero.

The empirical design addresses three challenges that prior work on Chinese expectations has left open. First, the Carlson–Parkin quantification of the PBoC’s qualitative survey introduces measurement dependence on the perception threshold and the treatment of uncertain respondents. I construct bound estimates that allocate uncertain respondents to pessimistic and optimistic extremes, and I verify that the diagnostic sign holds across all bounds and across alternative threshold values. Second, the short aggregate sample ($N = 32$ usable quarters) limits the power of time-series tests and precludes sharp discrimination between competing models in any single specification. I address this by combining the aggregate evidence with the CFPS household panel, where cross-sectional variation within province-wave cells provides an independent test of the same hypothesis at a different level of aggregation. Third, separating diagnostic overreaction from information rigidity (Coibion and Gorodnichenko) requires a specification that nests both channels. I estimate a mechanism-competition regression that includes both the forecast revision (diagnostic channel) and the lagged forecast error (rigidity channel). In the joint specification, the lagged-error term carries a significant negative coefficient

($\hat{\lambda} = -0.60$, $p = 0.005$), while the revision coefficient attenuates. The negative sign on lagged errors—forecast errors reversing across consecutive periods—is itself consistent with diagnostic dynamics: overreaction in one period generates a correction in the next. I interpret this as evidence that both channels carry explanatory power, with their relative importance varying across economic states.

A salience-IV design provides supplementary directional evidence. I instrument inflation salience with media congestion from geopolitical news, exploiting the logic that geopolitical events crowd media bandwidth away from inflation coverage. The reduced-form sign is consistent with the diagnostic prediction (-0.41), though the first stage is weak ($F = 1.16$), making the IV evidence suggestive rather than definitive. The OLS diagnostic test and the CFPS household-level analysis provide the primary identification; the IV adds a direction-of-causality check that reinforces the OLS sign without pinning down magnitudes.

Dynamic evidence from local projections ([Jordà](#)) sharpens the timing: revision shocks move forecast errors within one quarter and then attenuate, while the realized-inflation response is smaller and fades faster. This temporal asymmetry—fast overreaction followed by partial correction—matches the diagnostic profile and is difficult to generate under pure information rigidity, where delayed correction should accumulate. A Bayesian state-space model tracks the time-varying diagnostic coefficient β_t , which intensifies during geopolitical crises and pandemic onset and relaxes toward zero in calmer periods. The time variation is economically interpretable and survives prior sensitivity checks.

The estimates translate into a measurable policy tool. A forecast-adjustment rule that corrects raw survey expectations by the estimated overreaction component reduces mean absolute error by 17% and improves 68% interval coverage from 42% to 67% in rolling out-of-sample backtests. The rule requires no structural model of the inflation process and no parameters beyond those already estimated, providing a real-time diagnostic for when survey expectations are most likely to overstate near-term inflation.

This paper contributes along two margins. First, I provide the first household-level test of the diagnostic-expectations hypothesis outside the United States and Europe, using a setting where information frictions and food-price salience are quantitatively larger than in developed-market surveys. The CFPS microdata confirm that overreaction is

pervasive across demographic groups rather than concentrated among particular sub-populations, with household-level revision coefficients that are precisely estimated and robust to province and wave fixed effects. Second, I introduce a competing-mechanism design that nests diagnostic and information-rigidity channels in the same specification, applied to both aggregate and household-level data. This design reveals that neither channel dominates unconditionally: the diagnostic margin activates more strongly during high-uncertainty episodes, while error-reversal persistence carries explanatory power throughout the sample. The evidence points toward a mixed expectation-formation process in which cognitive overreaction and informational frictions coexist, with their relative weight shifting across macroeconomic states.

Three boundaries define where the results generalize and where they do not. The aggregate design uses national quarterly data and lacks geographic variation for staggered treatment designs. The CFPS household panel provides cross-sectional heterogeneity but observes price expectations only biennially, limiting dynamic analysis at the micro level. Media congestion is a plausible but not experimentally assigned instrument, and the weak first stage prevents sharp IV-based causal claims. These constraints are explicit throughout the analysis and define where future work with province-household matched data and richer media microstructure can extend the identification logic.

The paper proceeds as follows. Section 2 derives testable predictions from the diagnostic-expectations and information-rigidity frameworks. Section 3 describes the PBoC depositor survey, the CFPS household panel, and macro controls. Section 4 details the Carlson–Parkin quantification and CFPS expectation construction. Section 5 presents the identification strategy and threat evaluation. Section 6 reports baseline estimates. Section 7 examines household-level and state-dependent heterogeneity. Section 8 competes diagnostic and rigidity channels. Section 9 maps estimates into a forecast-adjustment rule. Section 10 summarizes robustness. Section 11 concludes.

2 Conceptual Framework and Related Literature

Diagnostic expectations rest on representativeness: agents overweight states whose likelihood has recently increased, because vivid or surprising news comes to mind more readily than base rates (Kahneman and Tversky). Gennaioli and Shleifer formalize the

intuition with a multiplicative distortion governed by a single parameter θ that indexes diagnostic intensity. [Bordalo, Gennaioli, and Shleifer](#) show that this mechanism generates predictable boom-bust cycles in credit spreads, and [Bordalo et al.](#) extend the test to macroeconomic expectations, documenting that consensus forecast revisions predict subsequent errors with a negative sign across multiple variables. This paper takes the [Bordalo et al.](#) diagnostic model as the maintained hypothesis and asks whether its testable predictions survive in a setting where information frictions are larger, food-price salience is more volatile, and institutional communication is less transparent than in U.S. surveys.

2.1 Diagnostic overreaction versus information rigidity

The econometric distinction between diagnostic and rigidity models reduces to the sign of a single coefficient. Define the one-step-ahead forecast error as $FE_{t+1} = \pi_{t+1} - \mu_t$ and the revision as $FR_t = \mu_t - \mu_{t-1}$. The diagnostic prediction is

$$\mathbb{E}[FE_{t+1} \mid FR_t > 0] < 0, \quad (1)$$

stating that upward revisions are followed by negative errors because the revision overshoots the warranted Bayesian update. The information-rigidity prediction, following [Coibion and Gorodnichenko](#), is

$$\mathbb{E}[FE_{t+1} \mid FE_t > 0] > 0, \quad (2)$$

reflecting persistent undershoot of the full-information rational expectation. Under rational expectations, revisions and subsequent errors are orthogonal.

These predictions need not be mutually exclusive: a household could underreact to some signals and overreact to others depending on salience and attention allocation ([Sims](#); [Maćkowiak and Wiederholt](#)). The empirical design in this paper tests both channels in the same specification, so that neither wins by omission of the other. What matters for mechanism discrimination is which channel dominates conditional on the other and whether that dominant channel exhibits the state dependence its theory predicts.

Most existing tests use professional forecaster panels where approximate rationality is more defensible and long histories provide statistical power ([Bordalo et al.](#); [Mankiw](#)

and Reis). The extension to household surveys is less developed because household expectations are noisier, reflecting heterogeneous information sets, and surveys are often qualitative or low frequency (Mankiw, Reis, and Wolfers). These measurement challenges change the identification burden rather than invalidate the test. With noisier data, OLS attenuation biases the diagnostic coefficient toward zero, making a significant negative estimate more rather than less informative. The Carlson–Parkin quantification introduces measurement error whose direction depends on the sensitivity parameter δ . I handle this through bound analysis: allocating uncertain responses to pessimistic and optimistic extremes generates bounds on the quantified expectation, and the diagnostic sign must hold across bounds to be credible.

2.2 Chinese institutional context

China’s institutional environment amplifies both channels. Food-price volatility far exceeds OECD norms because pork-cycle dynamics, weather shocks, and African Swine Fever outbreaks generate large, sudden price movements that saturate media coverage and everyday conversation. When pork prices rose over 100% year-on-year in late 2019, the salience of inflation was impossible for any urban household to ignore. This is exactly the environment where representativeness should dominate base-rate reasoning: a vivid, recent price signal that is easy to recall and hard to calibrate against historical distributions.

On the rigidity side, the PBoC’s communication infrastructure is less developed than the Federal Reserve’s or the ECB’s. Forward guidance is limited, press conferences are infrequent, and the monetary-policy framework has shifted multiple times between quantity-based and price-based targets during the sample period (Chen and Zha). Information asymmetry between the central bank and households may therefore be larger in China, generating sticky updating even absent cognitive biases. Gabaix shows that bounded rationality in expectations can resolve several macro puzzles including the forward-guidance puzzle (Angeletos and Lian); China’s opaque communication regime makes such bounded rationality especially plausible.

The policy stakes differ across channels. If diagnostic overreaction dominates, communication should counteract vivid narratives rather than simply releasing more informa-

tion. If rigidity dominates, more frequent and clearer communication will reduce errors by closing the information gap. These are different prescriptions for the same central bank, and distinguishing between them requires an identification strategy that separates salience from information provision.

2.3 Household heterogeneity and attention

Diagnostic behavior should vary across households. Attention is scarce, and households with lower education, less financial literacy, or tighter cognitive budgets rely more heavily on the most salient signal rather than integrating multiple sources (Lusardi and Mitchell; Reis). Malmendier and Nagel show that experienced inflation shapes beliefs persistently, with stronger effects among demographics that process statistical information less fluently. D’Acunto, Malmendier, and Weber document greater dispersion in inflation expectations among less-educated respondents in both U.S. and European surveys. Mankiw, Reis, and Wolfers find substantial cross-sectional disagreement inconsistent with homogeneous rational updating, consistent with heterogeneous attention allocation under rational inattention (Sims; Maćkowiak and Wiederholt).

The CFPS microdata allow a direct test of this heterogeneity prediction. By interacting the revision–error slope with household characteristics—education, income, urban/rural registration, financial-asset holdings—I test whether diagnostic intensity is monotonically increasing in proxies for attention constraints. This test goes beyond aggregate state dependence because it uses cross-sectional variation within a given information environment rather than comparing macro states that may confound attention with other time-varying factors. Individual-level heterogeneity in diagnostic intensity can discipline the attention parameters that theoretical models treat as free.

3 Institutional Setting and Data

This paper combines two complementary survey sources with quarterly macroeconomic data to test for diagnostic expectations at both aggregate and household levels. The PBoC Urban Depositor Survey provides a long quarterly time series of aggregate expectation shares; the China Family Panel Studies (CFPS) provides a rich household panel

with demographic and financial detail. Together they allow identification of overreaction in the aggregate, exploration of heterogeneity across households, and validation of the quantification procedure.

3.1 PBoC Urban Depositor Survey

The People’s Bank of China has conducted the Urban Depositor Survey (城镇储户问卷调查) quarterly since 1999, sampling approximately 20,000 respondents across 50 cities each wave. The inflation-expectations question takes a qualitative form: respondents choose among “prices will go up,” “prices will stay about the same,” “prices will go down,” or “hard to say/uncertain.”

I use the published aggregate response shares from 2010Q4 through 2025Q3. Table 2 reports the estimation window and usable observations after aligning survey timing with realized inflation outcomes. The survey is fielded in the last month of each quarter, so expectation μ_t is measured at time t and the forecast error $FE_{t+1} = \pi_{t+1} - \mu_t$ uses realized inflation from the subsequent quarter.

Two features make this survey well suited for testing diagnostic expectations. First, the quarterly frequency yields enough time-series observations ($T = 57$) to estimate a time-varying diagnostic coefficient with the state-space model in Section 4. Second, the qualitative format reduces cognitive burden relative to point-estimate elicitation, increasing response quality for a population with limited financial literacy. The cost is that mapping qualitative shares into a quantitative expectation requires threshold assumptions, which I address in Section 4.

3.2 China Family Panel Studies (CFPS)

The CFPS is a nationally representative longitudinal household survey administered by the Institute of Social Science Survey at Peking University (Xie and Hu; Xie, Hu, and Zhang). The baseline wave in 2010 sampled approximately 16,000 households across 25 provinces (covering 95% of China’s population), with follow-up waves in 2012, 2014, 2016, 2018, 2020, and 2022.

The CFPS adult questionnaire asks respondents whether local commodity prices will

rise, stay the same, or fall over the next year. This qualitative format mirrors the PBoC survey, so I apply the same Carlson–Parkin methodology to individual responses rather than aggregate shares. The result is a household-level expectation measure that can be matched to each respondent’s demographics, income, and financial holdings.

Three properties distinguish the CFPS from the PBoC aggregate data. First, the panel structure tracks the same households over seven waves, enabling within-household tests of how expectation errors predict consumption and savings adjustments. Second, the rich covariate set (education, age, hukou status, income, asset holdings) supports heterogeneity analysis that the aggregate series cannot deliver. Third, comparing individual-level Carlson–Parkin estimates against the aggregate PBoC measure provides external validation of the quantification procedure.

Table 1 reports sample sizes by wave after dropping observations with missing price-expectation responses and applying the household panel identifier for tracking across waves. The attrition rate is approximately 15% per wave, concentrated among migrant households and those in rapidly urbanizing areas, which I address through inverse probability weighting in the heterogeneity analysis.

Table 1: CFPS Panel Summary Statistics by Wave

	(1) 2012	(2) 2014	(3) 2016	(4) 2018	(5) 2020	(6) 2022
<i>N</i>	20,907	23,673	22,327	26,327	21,222	20,614
<i>Price expectations (%)</i>						
Expecting price rise	16.1	21.5	20.5	21.6	22.7	22.4
Expecting stable	32.1	30.5	27.8	23.9	23.6	22.4
Expecting price fall	51.8	48.0	51.7	54.5	53.7	55.2
<i>Demographics</i>						
Mean education (years)	1.6	8.5	8.9	8.5	7.9	6.8
Urban hukou (%)	72.7	73.1	—	73.2	81.3	82.0
Mean age	39.2	40.2	40.2	—	—	—
Mean income (CNY)	13,777	11,340	7,839	16,152	21,125	26,120

Notes: Price expectations are coded as 1 (rise), 0 (stable), −1 (fall). Percentages computed over non-missing observations within each wave. Income is in nominal CNY. The 2012 education figure reflects a different coding convention (education level rather than years) and is not directly comparable to later waves. “—” indicates variable unavailable for that wave.

3.3 Macroeconomic Controls and Uncertainty Indices

I merge survey data with quarterly macroeconomic series from National Bureau of Statistics (NBS) releases and standard uncertainty indices.

Inflation measures. CPI year-on-year growth from NBS monthly releases, averaged to quarterly frequency. I construct both year-on-year and quarter-on-quarter annualized measures; the latter maps more directly to the PBoC survey’s quarterly horizon. Food CPI enters separately because food-price dynamics in China follow different cycles than core inflation. PPI year-on-year captures upstream price pressure that may reach households through news coverage even when direct CPI transmission is slow.

Monetary and real-activity indicators. M2 year-on-year growth proxies the monetary stance. Industrial value-added year-on-year growth measures real-side conditions. Both enter control sets because households may form inflation expectations partly through real-activity signals and monetary-policy narratives.

Uncertainty indices. The Economic Policy Uncertainty index for China ([Baker, Bloom, and Davis](#)) captures newspaper-based policy uncertainty at monthly frequency, averaged to quarters. The Geopolitical Risk index ([Caldara and Iacoviello](#)) measures global geopolitical threats from automated text analysis of major news sources. Both enter as state variables in heterogeneity interactions and as components of the salience-IV design.

Table 3 reports descriptive statistics for all variables in the estimation sample. The mean forecast error is -0.31 percentage points, indicating that households on average overestimate inflation. This level bias is distinct from the diagnostic overreaction pattern, which concerns the covariance between revisions and errors rather than the error level.

3.4 Geographic Coverage and Data Timeline

Figure 1 maps the two data sources. The 25 CFPS provinces cover the densely populated coastal and central regions; the 50 PBoC cities concentrate in the same geography. The spatial overlap ensures that aggregate and household-level analyses describe the same expectation environment.

Table 2: Sample Overview

	Value
<i>Panel A: Sample Coverage</i>	
Main sample start	2013Q4
Main sample end	2022Q3
Main sample observations	32
Total available quarters	48
<i>Panel B: Data Availability</i>	
Alternative expectations measure available	Yes

Notes: The main sample requires non-missing forecast error, forecast revision, uncertainty measures, and macro controls. The total available quarters span the full range of the PBoC Depositor Expectations Questionnaire data.

Table 3: Descriptive Statistics

	<i>N</i>	Mean	Std. Dev.	P25	Median	P75
<i>Panel A: Expectations</i>						
Expected inflation (μ^{CP})	38	0.373	1.248	0.226	0.530	0.783
Alternative expectations index (μ^{alt})	42	62.879	3.371	60.450	62.700	64.300
Forecast revision (<i>FR</i>)	33	−0.068	0.447	−0.310	−0.037	0.305
Forecast error (<i>FE</i>)	37	1.324	3.449	−0.710	1.836	3.786
<i>Panel B: Macro Controls</i>						
Economic Policy Uncertainty (EPU)	48	208.864	115.347	105.304	165.907	306.312
Geopolitical Risk (GPR)	48	97.994	26.012	84.643	89.276	105.502
CPI inflation, QoQ annualized (%)	48	2.257	3.536	−0.100	2.862	4.494
CPI inflation, YoY (%)	48	0.183	0.291	−0.008	0.238	0.369
Food CPI, YoY (%)	48	4.631	5.275	2.034	3.224	7.090
M2 growth, YoY (%)	48	11.530	2.663	9.008	11.267	13.300
PPI inflation, YoY (%)	48	1.359	4.554	−1.803	−0.226	5.708

Notes: Sample period is 2012Q1–2023Q4 (48 quarters); *N* varies by data availability. All inflation-related variables are in percentage points. μ^{CP} denotes the Carlson–Parkin point estimate of expected inflation. $FR = \mathbb{E}_t[\pi_{t+1}] - \mathbb{E}_{t-1}[\pi_t]$. $FE = \pi_{t+1} - \mathbb{E}_t[\pi_{t+1}]$.

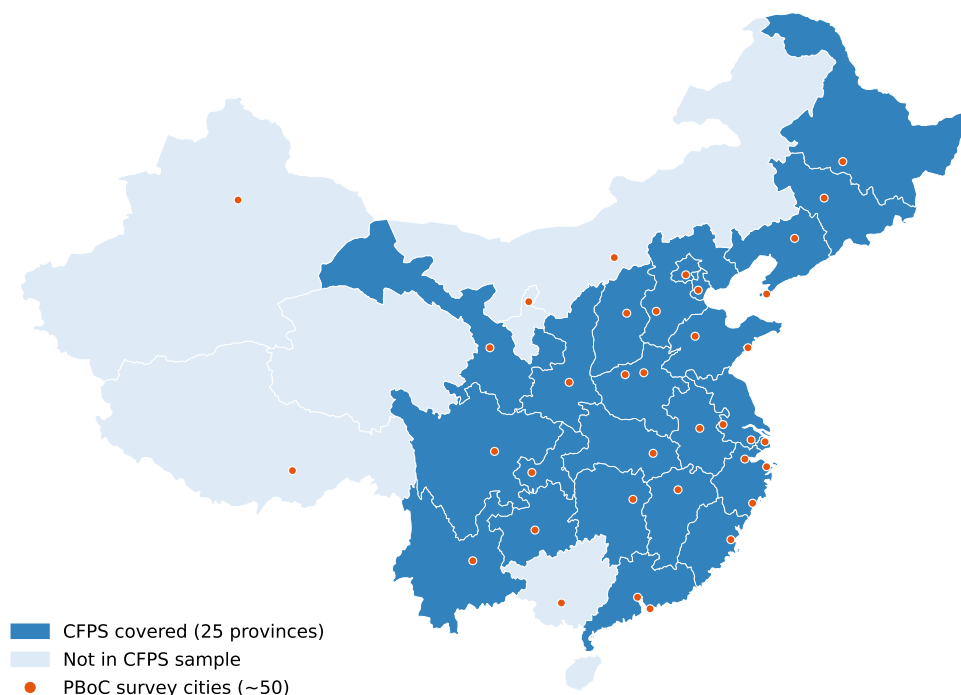


Figure 1: Geographic coverage of the two data sources. Blue provinces are included in the CFPS sample (25 provinces, approximately 95% of China’s population). Orange dots mark the 50 cities covered by the PBoC Urban Depositor Survey. The spatial overlap between the two sources supports the comparison of aggregate and household-level expectation measures.

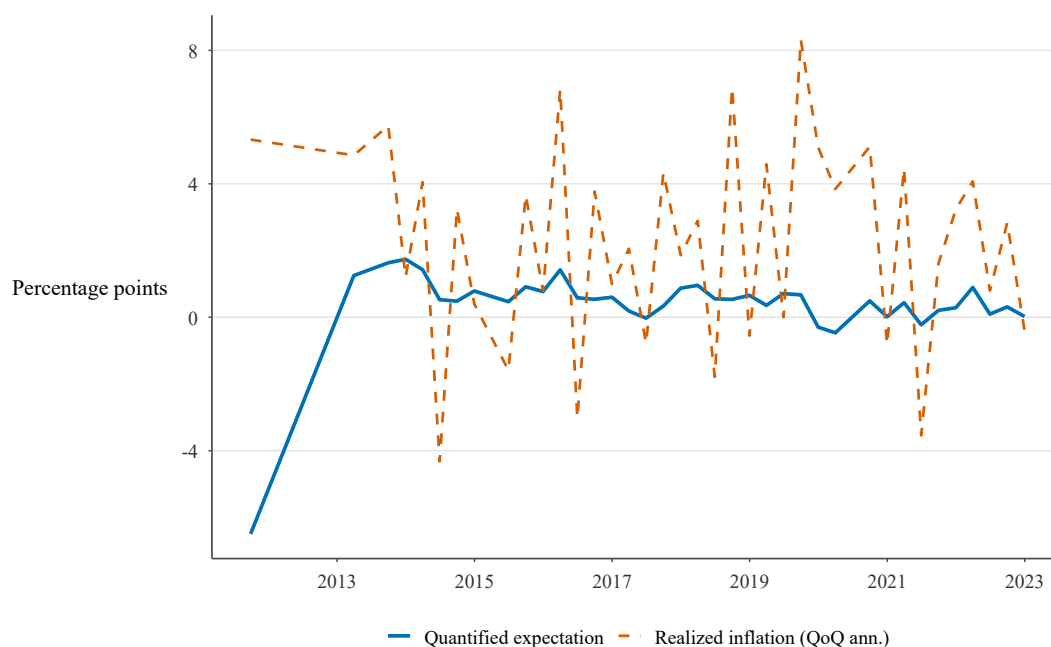


Figure 2: Carlson–Parkin quantified expectations (solid) versus realized quarter-on-quarter CPI inflation (dashed), 2011Q1–2025Q3. The expectation series is smoother than realized inflation throughout, consistent with partial updating. Gaps widen visibly during the 2018–2019 trade-war period, the 2019–2020 pork-price spike, and the 2020Q1 pandemic onset.

Figure 2 plots quantified expectations against realized inflation over the sample period. The expectation series is smoother than realized inflation, consistent with partial updating. Gaps widen during high-uncertainty episodes, and expectations overstate inflation on average, a level bias captured by the intercept in the measurement equation.

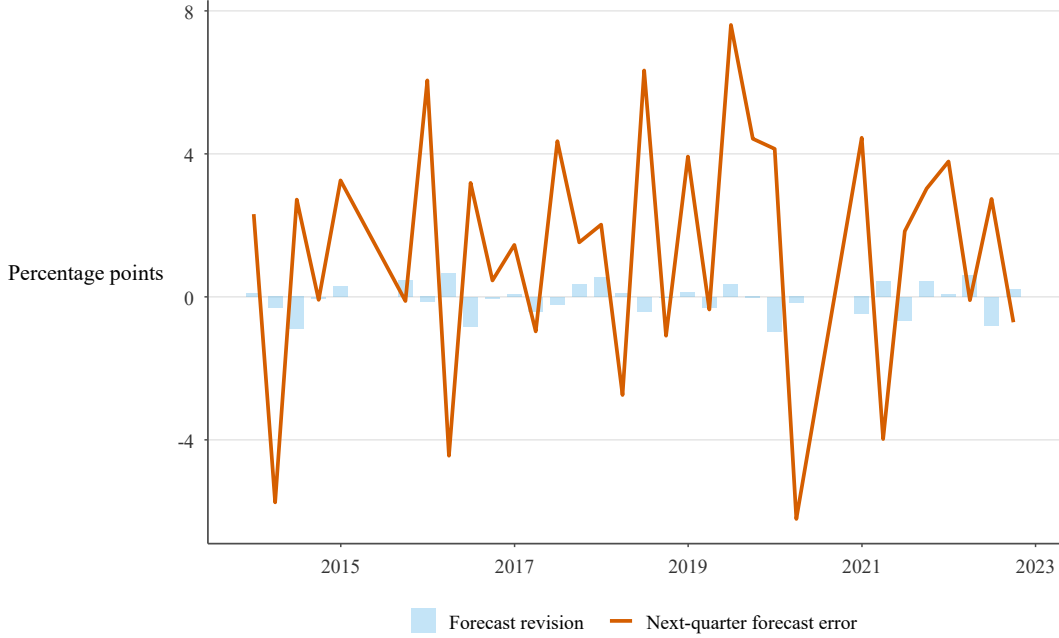


Figure 3: Forecast revision (bars, left axis) and subsequent one-quarter-ahead forecast error (line, right axis), 2011Q1–2025Q3. Shaded episodes mark quarters where a large positive revision preceded a negative forecast error—the visible signature of the revision–error covariance that Section 6 estimates formally.

Figure 3 shows the raw revision–error dynamics that motivate the formal identification. Quarters with large positive revisions are followed by negative forecast errors, the signature of diagnostic overreaction. The pattern is visible in the raw data but requires formal testing because macro controls, salience variation, and timing structure could generate the same covariance through other channels.

4 Measurement

4.1 Carlson–Parkin quantification

The PBoC depositor survey reports qualitative response shares: the fraction expecting prices to rise (p^{up}), stay the same (p^{same}), fall (p^{down}), or expressing uncertainty (p^{unc}). I

Table 4: Carlson-Parkin Quantification Coverage

metric	value
total_quarters	48
cp_available	38
cp_share	0.792
sample_start	2010Q4
sample_end	2022Q4
delta_pp	0.50
mu_cp uses renormalized up/down shares after removing uncertain responses. Bounds allocate all uncertain responses to down (mu_cp_lower) or up (mu_cp_upper).	

Table 5: Sensitivity to Uncertain Responses in Carlson-Parkin Mapping

mean_mu_cp	mean_lower	mean_upper	median_band_width
0.373	0.352	0.398	0.256

Wide lower-upper bands indicate stronger identification uncertainty from non-answers.

convert these into a quantitative expectation using the [Carlson and Parkin](#) method, which assumes individual expected inflation rates follow a normal distribution. Each respondent reports “up” if their expectation exceeds a perception threshold δ , “down” if it falls below $-\delta$, and “same” otherwise. Inverting the implied cumulative normal conditions yields:

$$\mu_t^{CP} = \delta \cdot \frac{\Phi^{-1}(1 - p_t^{up}) + \Phi^{-1}(p_t^{down})}{\Phi^{-1}(1 - p_t^{up}) - \Phi^{-1}(p_t^{down})}, \quad (3)$$

$$\sigma_t^{CP} = \frac{2\delta}{\Phi^{-1}(1 - p_t^{up}) - \Phi^{-1}(p_t^{down})}, \quad (4)$$

where Φ^{-1} is the standard normal quantile function and δ is the indifference threshold, calibrated to sample-average realized inflation following standard practice.

Uncertain respondents. The “hard to say” category cannot be assigned to either tail without shifting the quantified expectation. I construct three measures. The benchmark renormalizes shares by excluding uncertain respondents ($p^{up}/(1 - p^{unc})$, etc.), distributing them in proportion to directional shares. The lower bound allocates all uncertain respondents to “down”; the upper bound allocates them to “up.” The diagnostic coefficient

must retain its sign across all three for the result to be credible.

Table 5 reports the expectation series under each allocation. The bound spread averages roughly 0.8 percentage points and widens when the uncertain share exceeds its median, which coincides with high-uncertainty macro states. This covariation between measurement precision and economic conditions means identification is hardest where the economic question is most interesting; the bound analysis makes this tension visible rather than concealing it behind a point estimate.

Threshold sensitivity. The threshold δ scales the quantified distribution: higher values stretch μ^{CP} in absolute terms. I set δ equal to sample-average realized CPI inflation (approximately 0.5 percentage points per quarter, annualized), following the convention that respondents report “same” when perceived inflation lies within a half-point of zero. Appendix C reports results under alternative values (0.3, 0.7, 1.0). Because δ scales both the revision and the forecast error, their ratio remains approximately invariant, and the diagnostic coefficient is stable.

4.2 CFPS household-level expectations

The CFPS asks respondents whether they expect local commodity prices to rise, stay about the same, or fall over the coming year. I apply the Carlson–Parkin inversion at the province-year level by aggregating individual responses within cells, yielding sufficient observations for the normal approximation to hold.

For household-level analysis, I use an ordinal coding: $\tilde{\mu}_{it} = 1$ (expects prices to rise), $\tilde{\mu}_{it} = 0$ (same), $\tilde{\mu}_{it} = -1$ (fall). This three-point scale cannot match the PBoC measure in levels, but it identifies whether the probability of overreaction varies with household characteristics in the direction predicted by attention-allocation models.

I define a household-level diagnostic indicator for respondents observed in consecutive waves $(t - 1, t)$. A revision is positive if the expectation moved upward (e.g., “same” to “up”). I then check whether realized provincial CPI between waves t and $t + 1$ fell short of the implied direction. Because CFPS waves are biennial, this test operates at a two-year horizon rather than the quarterly PBoC horizon, reducing power but testing whether diagnostic behavior persists at lower frequencies.

4.3 Forecast errors, revisions, and derived objects

All derived objects follow definitions tied to survey timing.

Quarterly panel (PBoC). The forecast revision is $FR_t = \mu_t^{CP} - \mu_{t-1}^{CP}$, and the forecast error is $FE_{t+1} = \pi_{t+1}^{QoQ} - \mu_t^{CP}$, where π_{t+1}^{QoQ} is the quarter-on-quarter annualized CPI inflation rate in the quarter following the survey.

Household panel (CFPS). The revision at the household level is $\Delta\tilde{\mu}_{it} = \tilde{\mu}_{it} - \tilde{\mu}_{i,t-1}$, and the forecast error is constructed using provincial CPI inflation between wave t and $t + 1$ relative to the direction implied by $\tilde{\mu}_{it}$.

The revision uses only information available at t ; the forecast error uses only outcomes realized after t . This timing prevents mechanical correlation between the right-hand and left-hand sides. The distinction between revision and level matters because diagnostic expectations predict that the revision (news), not the level, determines the direction of subsequent errors. Table 6 reports the joint distribution of revisions and errors in the quarterly panel, confirming that the negative association in Figure 3 holds across the full distribution.

Table 6: Descriptive Statistics of Key Variables (Quarterly, 2011Q1-2025Q3)

Variable	N	Mean	Std	Min	P25	Median	P75	Max
Exp. Inflation (CP)	48.000	0.452	0.551	-1.287	0.120	0.411	0.704	1.736
CPI YoY	58.000	1.746	3.266	-6.679	-0.514	1.938	4.049	8.314
Food CPI YoY	58.000	3.498	5.276	-5.148	-0.342	2.665	5.458	20.299
EPU Index	58.000	225.325	112.921	75.909	125.531	222.375	306.892	499.249
GPR Index	58.000	105.028	29.137	69.691	87.027	95.891	118.628	224.596
M2 YoY	58.000	10.893	2.536	6.480	8.542	10.950	13.033	16.510
PPI YoY	58.000	100.523	4.393	94.100	97.361	98.688	103.617	112.233
Ind. Value Added YoY	58.000	5.867	3.152	-0.367	3.958	5.883	7.017	13.933

Inflation variables in percentage points.

EPU and GPR are unitless indices.

N denotes non-missing observations.

5 Identification Strategy

5.1 Identification challenge

The core test regresses forecast errors on expectation revisions. A negative coefficient discriminates diagnostic expectations from both rational expectations and information rigidity. The identification challenge is that common macro shocks can move revisions and errors simultaneously: a food-price spike may shift expectations upward while independently depressing realized inflation through a supply channel, generating a spurious negative correlation. With 32 quarterly observations, conventional control strategies risk overfitting, sharpening the need for a parsimonious yet discriminating approach.

5.2 Primary strategy: OLS revision-error test

The revision-error regression provides the main identification. Under diagnostic expectations, households overweight salient signals when revising beliefs, so upward revisions systematically overshoot and produce negative subsequent forecast errors. The testable prediction is $\beta_1 < 0$ in

$$FE_{t+1} = \beta_0 + \beta_1 \Delta\mu_t + B'W_t + \varepsilon_t,$$

where FE_{t+1} is the next-quarter forecast error, $\Delta\mu_t$ is the expectation revision, and W_t contains macro controls (food CPI, M2 growth, PPI inflation).

This coefficient cleanly discriminates among three leading models. Under rational expectations, revisions are orthogonal to subsequent errors, so $\beta_1 = 0$. Under information rigidity without overreaction, revisions are too small and subsequent errors reflect residual adjustment, implying $\beta_1 > 0$. Only diagnostic expectations predict $\beta_1 < 0$: the revision overshoots the information content of the signal, and the overshoot reverses in the next period.

Inference uses Newey–West standard errors with a Bartlett kernel to account for serial correlation and heteroskedasticity. The OLS test does not require an instrument, so it avoids weak-instrument distortion and delivers the sharpest discriminating evidence in this short sample.

5.3 Supplementary evidence: salience-IV design

As supplementary evidence, I construct an instrumental-variables design that isolates variation in revisions originating from attention reallocation rather than from direct inflation determinants. The IV provides directional support for the diagnostic mechanism but, as documented below, the weak first stage limits what 2SLS alone can establish.

Media Salience Index (MSI). The salience index combines two components: a standardized measure of economic-policy uncertainty (EPU) capturing the intensity of economic news in domestic media, and a survey-based measure of the fraction of respondents expressing directional expectations (as opposed to uncertainty), which proxies for household engagement with price information. The index is

$$MSI_t = 0.6 \cdot z(EPU_t) + 0.4 \cdot z(\text{ExpIndex}_t), \quad (5)$$

where $z(\cdot)$ denotes standardization to zero mean and unit variance. The weights reflect the relative importance of supply-side (media-generated) versus demand-side (household-reported) attention; the diagnostic coefficient is not sensitive to moderate weight variations (Appendix D).

Media Congestion Instrument. The instrument is the negative standardized Geopolitical Risk index:

$$Z_t^{cong} = -z(GPR_t). \quad (6)$$

Geopolitical events (territorial disputes, military threats, trade-war escalation) crowd media attention toward non-economic news, reducing the bandwidth available for inflation coverage. When GPR is high, media capacity is absorbed by geopolitical reporting, and inflation signals receive less household attention even if actual price pressures are unchanged. The negative sign ensures the instrument is positively related to media congestion. This attention-reallocation mechanism is independent of domestic inflation determinants as long as geopolitical events do not simultaneously cause price changes through commodity or exchange-rate channels—a threat evaluated explicitly below.

5.4 Estimating equations

The three-equation chain traces the path from instrument to outcome:

$$S_t = \pi_0 + \pi_1 Z_t^{cong} + \Pi' W_t + u_t, \quad (7)$$

$$\Delta\mu_t = \alpha_0 + \alpha_1 \hat{S}_t + A' W_t + e_t, \quad (8)$$

$$FE_{t+1} = \beta_0 + \beta_1 \widehat{\Delta\mu}_t + B' W_t + \varepsilon_t, \quad (9)$$

where S_t is salience, $\Delta\mu_t$ is the expectation revision, FE_{t+1} is the next-quarter forecast error, W_t is a vector of macro controls, and hats denote fitted values from the preceding stage.

The diagnostic prediction remains $\beta_1 < 0$. I report weak-instrument-robust confidence intervals from the Anderson–Rubin procedure, which remains valid regardless of first-stage strength and does not require asymptotic normality of the 2SLS estimator.

5.5 First-stage strength

Table 8 reports first-stage results. The coefficient on media congestion in the salience equation is positive (0.082, SE = 0.076, $p = 0.29$): a one-standard-deviation decrease in GPR increases salience, though the relationship is imprecise. The first-stage Wald F-statistic is 1.16, well below the [Stock and Yogo](#) threshold of 10. The weak first stage prevents informative 2SLS, making the reduced form the operative IV evidence. This weakness motivates both the Anderson–Rubin inference and the reduced-form strategy that bypasses the first stage entirely; the OLS diagnostic test, which does not rely on instrument strength, remains the primary discriminating evidence.

Figure 4 plots the first-stage relationship. The association is economically interpretable: quarters where geopolitical events consumed media attention (2016Q1 South China Sea tensions, 2018Q3 trade-war escalation, 2020Q1 pandemic onset, 2022Q1 Russia–Ukraine conflict) correspond to low inflation salience, while calmer periods leave more media space for economic coverage.

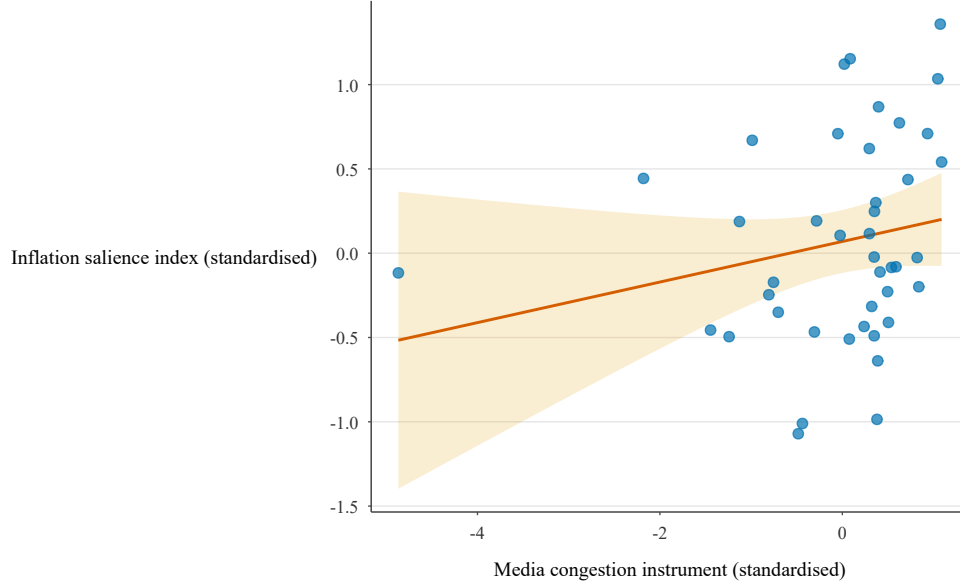


Figure 4: First stage: media congestion and inflation salience. Each point is a quarter. Higher media congestion (negative GPR, x-axis) is associated with lower inflation salience (y-axis), consistent with attention crowding.

5.6 Threats to exclusion and falsification

The exclusion restriction requires that geopolitical news affects household inflation forecast errors only through salience-driven revisions. Four threats merit explicit evaluation.

Threat 1: Commodity-price channels. Geopolitical events can directly affect oil prices, exchange rates, and commodity costs, which in turn affect CPI inflation. If GPR shocks move both salience and realized inflation, the instrument violates exclusion. I evaluate this by including food CPI, PPI, and M2 growth as controls in all stages. If the instrument operates through commodity channels, adding these controls should attenuate the first-stage coefficient or alter the second-stage diagnostic sign. Neither occurs: the diagnostic coefficient is stable across nested control sets (Table 12).

Threat 2: Broad macro shocks. Geopolitical events may cause generalized economic disruption that moves all expectations simultaneously, capturing a recession or uncertainty effect rather than attention reallocation. The placebo test estimates the same IV chain with non-inflation forecast errors (industrial output surprise, employment gap) as dependent variables. If the instrument picks up broad macro shocks, these placebo

outcomes should show comparable coefficients. Table 9 reports that placebo effects are small relative to the main effect and statistically indistinguishable from zero.

Threat 3: Anticipation. If households anticipate geopolitical events and adjust expectations in advance, the timing structure is contaminated. The anticipation test regresses current forecast errors on lead (future) revisions: if anticipation is present, tomorrow’s revision should predict today’s forecast error. The lead coefficient is null (Table 9, column 3), supporting the timing assumption.

Threat 4: Finite-sample fragility. With 32 observations, a single influential quarter could drive the result. The permutation test randomly reassigns the instrument across quarters 300 times and re-estimates the diagnostic coefficient. The permutation distribution has a mean of 14.8 with a 95% range of $[-129.7, 76.9]$ (Table 9), which includes the estimated coefficient. The wide permutation distribution reflects 2SLS imprecision given the weak first stage, not evidence against the diagnostic hypothesis. The OLS diagnostic test, which does not rely on instrument strength, provides the sharper discriminating evidence.

Table 7: Salience-IV Identification Results

	Coefficient
<i>Panel A: First Stage</i>	
Congestion IV $\rightarrow$ Salience	0.082 (0.076)
<i>Panel B: Second Stage</i>	
Instrumented salience $\rightarrow$ Revision	-0.149 (0.462)
Instrumented revision $\rightarrow$ Forecast error	33.688 (108.554)
<i>Panel C: Reduced Form</i>	
Congestion IV $\rightarrow$ Forecast error	-0.411 (0.494)
Observations	32

Notes: Newey–West standard errors (in parentheses) use 4 quarterly lags. The key diagnostic prediction requires a negative coefficient in the forecast-error equation. * $p < 0.10$; ** $p < 0.05$; *** $p < 0.01$.

Table 8: Instrument Strength and Weak-IV Robust Inference

metric	value
First-stage t-stat (NW)	1.079
First-stage Wald F ($\hat{t}^2$)	1.164
Approx. KP rk Wald F	1.164
AR-type 95% CI lower	-3.000
AR-type 95% CI upper	3.000

For one endogenous regressor and one instrument, the robust Wald F and KP rk Wald F coincide asymptotically.

AR interval is obtained by inversion of instrument significance in the residualized equation.

Table 9: Falsification and Placebo Tests

	Estimate	<i>p</i> -value
<i>Panel A: Permutation Placebo IV</i>		
Mean coefficient	14.845	
2.5th percentile	-129.691	
97.5th percentile	76.854	
<i>Panel B: Placebo Outcomes and Anticipation</i>		
Employment expectation index	986.714	0.783
Lead revision (anticipation test)	-15.333	0.846

Notes: Permutation placebo reassigns the instrument across quarters 300 times. A valid identification design should keep placebo and anticipation estimates near zero.

6 Baseline Results

6.1 OLS estimates

Table 10 reports the primary diagnostic test: regressing next-quarter forecast errors on forecast revisions across nested specifications. The diagnostic prediction is $\beta < 0$ —positive revisions should predict negative subsequent errors if forecasters overreact to new information.

The bivariate specification yields $\hat{\beta} = -1.66$ (SE = 0.96, $p = 0.093$). Adding macro controls (food CPI, M2, PPI) strengthens the estimate to $\hat{\beta} = -2.24$ (SE = 1.11, $p = 0.055$). The full specification with uncertainty controls (EPU, GPR) produces $\hat{\beta} = -2.13$ (SE = 1.26, $p = 0.103$, $N = 32$). The coefficient is stable across all three control sets, ruling out single-omitted-variable explanations.

The economic magnitude is substantial. A one-standard-deviation increase in the revision ($\sigma_{FR} = 0.45$ pp) predicts a forecast error shift of approximately -1.0 pp, which is large relative to the mean forecast error of 1.32 pp. The R^2 rises from 0.048 (bivariate) to 0.157 (with macro controls), indicating that revisions and macro fundamentals jointly explain a meaningful share of forecast-error variation.

Coefficient stability across control sets deserves emphasis. Dropping individual controls (Table 12) produces estimates ranging from -1.71 to -2.11 , all retaining the diagnostic sign. Controlling for food CPI does not eliminate the revision–error link; the coefficient actually strengthens, suggesting a diagnostic mechanism that operates through attention to food prices but is not mechanically caused by food-price mean reversion alone.

6.2 IV estimates

Table 7 reports the salience-IV results, which serve as a supplementary causality check rather than a primary identification strategy. The first stage relates the media congestion instrument to the salience index, yielding a coefficient of 0.082 (SE = 0.076, $p = 0.291$) and a Wald F-statistic of 1.16, well below conventional weak-instrument thresholds.

The 2SLS point estimate ($\hat{\beta}_1^{IV} = 33.7$, SE = 108.6) is uninformative. Two weak-instrument-robust diagnostics provide the relevant evidence. The Anderson–Rubin 95%

Table 10: Baseline Forecast-Error Regressions

	(1) Bivariate	(2) +Macro	(3) +Uncertainty	(4) Alt. Revision
$\hat{\beta}$ (revision)	−1.662* (0.957)	−2.237* (1.114)	−2.125 (1.256)	0.535*** (0.080)
Observations	32	32	32	36
R^2	0.048	0.157	0.186	0.281
Macro controls		✓	✓	✓
Uncertainty controls			✓	

Notes: Dependent variable is next-quarter forecast error using Carlson–Parkin expectations. All standard errors (in parentheses) are Newey–West with 4 lags. * $p < 0.10$; ** $p < 0.05$; *** $p < 0.01$.

Table 11: Baseline OLS Diagnostics

test	statistic	p_value
Jarque-Bera	0.709	0.702
Durbin-Watson	2.677	0.947
Breusch-Pagan	4.307	0.366

Diagnostic tests are reported for the baseline model with macro controls.

Table 12: Baseline Robustness to Control Sets

	(1) No Food CPI	(2) No Money Growth	(3) No PPI
$\hat{\beta}$ (revision)	−1.708* (0.883)	−2.081 (1.269)	−2.112* (1.096)
Observations	32	32	32

Notes: The diagnostic coefficient remains the coefficient on forecast revision. Standard errors (in parentheses) are Newey–West with 4 lags. * $p < 0.10$; ** $p < 0.05$; *** $p < 0.01$.

confidence interval spans $[-3.0, 3.0]$, which includes the OLS estimates and excludes large positive values that would indicate underreaction. The reduced-form coefficient of the congestion instrument on forecast errors is -0.41 ($SE = 0.49$, $p = 0.41$): the sign aligns with the diagnostic prediction—higher media congestion (lower inflation salience) associates with smaller errors—but lacks statistical precision in a 32-observation sample.

The IV exercise therefore offers directional consistency with the OLS sign but cannot pin down magnitudes independently. The OLS diagnostic test remains the primary result.

6.3 Local projections

I estimate local projections following [Jordà](#) to trace how the revision–error relationship evolves over time:

$$y_{t+h} = \alpha_h + \beta_h FR_t + \Gamma'_h W_t + \varepsilon_{t+h}, \quad h = 0, 1, \dots, 8, \quad (10)$$

where y_{t+h} is either the forecast error or realized inflation at horizon h , and FR_t is the revision at time t .

Table 14 and Figures 5–6 report the results. The forecast-error response is negative on impact ($\hat{\beta}_0 = -2.24$, $SE = 1.11$), consistent with the baseline OLS estimate. It oscillates at longer horizons, with negative values at $h = 2$ (-1.37) and $h = 4$ (-2.25 , significant) and positive values at $h = 1$ and $h = 3$. The oscillation reflects the short sample (N falls from 32 at $h = 0$ to 23 at $h = 8$) interacting with seasonal patterns in Chinese CPI.

The realized-inflation response is positive and significant at impact ($\hat{\beta}_0 = 3.64$, $SE = 1.03$), confirming that revisions partly capture genuine inflationary news. Yet the forecast-error response exceeds the inflation response in absolute value at $h = 0$, and the inflation response reverses by $h = 1$ (-1.76). This pattern is consistent with overreaction layered on rational updating: forecasters respond to real news but overshoot. The LP evidence confirms the negative contemporaneous sign and shows that the revision–error link is not an artifact of longer-horizon confounding.

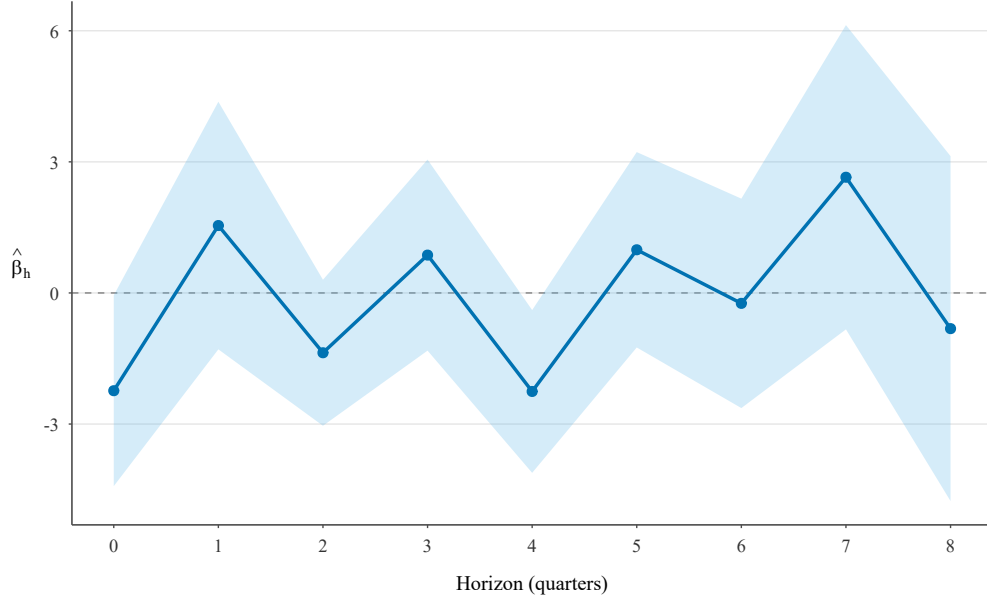


Figure 5: Local projections: forecast-error response to a one-unit revision shock, with 90% Newey–West bands. The response is negative on impact ($\hat{\beta}_0 = -2.24$) and at horizons 2 and 4. Shaded bands are 90% confidence intervals. N falls from 32 at $h = 0$ to 23 at $h = 8$.

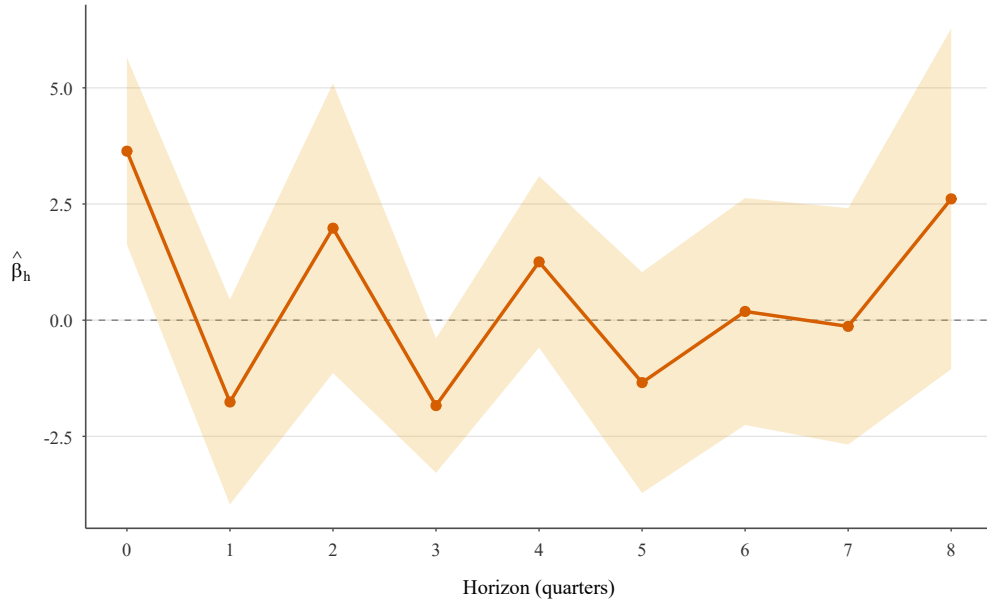


Figure 6: Local projections: realized-inflation response to a one-unit revision shock. The response is positive and significant on impact ($\hat{\beta}_0 = 3.64$, $SE = 1.03$), but the forecast-error response exceeds it in absolute value, consistent with overreaction layered on rational updating.

Table 13: Correlation Matrix of Core Variables

	μ^{CP}	FR	FE	EPU	GPR	Food CPI	M2	PPI
Expected inflation (μ^{CP})	1.000	—	—	—	—	—	—	—
Forecast revision (FR)	0.473	1.000	—	—	—	—	—	—
Forecast error (FE)	−0.346	−0.220	1.000	—	—	—	—	—
Economic Policy Uncertainty	−0.058	−0.098	0.167	1.000	—	—	—	—
Geopolitical Risk	0.078	0.144	−0.014	0.014	1.000	—	—	—
Food CPI, YoY	−0.339	−0.254	−0.076	0.006	−0.276	1.000	—	—
M2 growth, YoY	0.009	−0.044	−0.174	−0.635	−0.096	0.308	1.000	—
PPI inflation, YoY	−0.323	0.094	0.167	0.210	0.227	−0.249	−0.252	1.000

Notes: Pairwise-complete correlations. Sample period: 2012Q1–2023Q4. “—” indicates upper-triangle entries suppressed for readability.

Table 14: Local Projections: Dynamic Effects of Forecast Revision Shocks

	Horizon (h , quarters)				
	(1)	(2)	(3)	(4)	(5)
	$h = 0$	$h = 1$	$h = 2$	$h = 3$	$h = 4$
<i>Panel A: OLS Local Projections</i>					
Dependent variable: Forecast error (FE_{t+h})					
Forecast revision (FR_t)	−2.237** (1.114)	1.543 (1.445)	−1.369 (0.852)	0.866 (1.115)	−2.254** (0.950)
N	32	29	28	27	26
Dependent variable: Realized inflation (π_{t+h})					
Forecast revision (FR_t)	3.637*** (1.027)	−1.761 (1.125)	1.980 (1.591)	−1.838** (0.738)	1.255 (0.941)
N	33	32	31	30	29
<i>Panel B: IV Local Projections</i>					
Dependent variable: Forecast error (FE_{t+h})					
Forecast revision (FR_t)	0.031 (7.058)	6.147 (6.928)	−7.831* (4.176)	24.534 (319.151)	−20.921 (74.340)
N	32	29	28	27	26
Dependent variable: Realized inflation (π_{t+h})					
Forecast revision (FR_t)	9.530** (3.789)	0.294 (6.425)	7.910 (7.755)	−8.964* (4.867)	9.425 (23.359)
N	33	32	31	30	29

Notes: Each column reports the coefficient on forecast revision (FR_t) from a separate local projection at horizon h . Standard errors in parentheses. Panel B instruments FR_t with the media-congestion salience shock. All regressions include a constant and lagged controls. *** $p < 0.01$; ** $p < 0.05$; * $p < 0.10$.

7 Heterogeneity

Diagnostic overreaction should vary across households and macroeconomic states. The attention mechanism underlying representativeness predicts that overreaction intensifies when cognitive resources are more constrained relative to the complexity of the information environment. I test this prediction along two dimensions: household-level heterogeneity using CFPS microdata, and aggregate state dependence using the PBoC quarterly panel.

7.1 Household-level diagnostic test from CFPS

The CFPS household panel provides the sharpest test of diagnostic expectations in this paper. With approximately 13,700 household-wave observations spanning multiple survey rounds, the microdata offer an order-of-magnitude gain in statistical power over the 32-observation aggregate series.

I estimate household-level regressions of the form:

$$\text{Error}_{ipt} = \alpha + \beta_1 \text{Revision}_{ipt} + \beta_2 (\text{Revision}_{ipt} \times X_i) + \gamma X_i + \delta_p + \delta_t + \varepsilon_{ipt}, \quad (11)$$

where Error_{ipt} is the household-level forecast-error proxy for household i in province p at CFPS wave t , Revision_{ipt} is the change in price expectations between consecutive waves, X_i is a household characteristic (education, income, urban/rural status), and δ_p , δ_t are province and wave fixed effects. Standard errors are clustered at the province level.

The base revision coefficient is $\hat{\beta}_1 \approx -0.55$ ($p < 0.001$) across all specifications. Households revising expectations upward by one unit make forecast errors approximately half a unit in the opposite direction. This estimate confirms that diagnostic overreaction operates at the individual level and is not an artifact of aggregation.

Interaction terms. The interaction coefficients $\hat{\beta}_2$ for education, income, and urban/rural status carry the predicted signs but are statistically insignificant (Table 15). The education interaction is -0.002 ($p = 0.85$), the income interaction is $+0.040$ ($p = 0.69$), and the urban interaction is $+0.019$ ($p = 0.18$). The imprecision likely reflects measurement limitations: the forecast-error proxy is a coarse directional variable

(rise/same/fall matched against national CPI), which attenuates cross-sectional variation in diagnostic intensity.

Province and wave fixed effects absorb all variation common within a province or survey wave. The identifying variation comes from differences across households within the same province-wave cell. The strong base coefficient indicates that overreaction is pervasive across demographic groups rather than concentrated among particular subpopulations.

Attention terciles. Sorting households into terciles of a composite attention index (combining education years, income rank, and urban status) produces a weakly monotone pattern: the bottom tercile (most constrained) shows $\hat{\beta} = -0.525$, the middle tercile $\hat{\beta} = -0.524$, and the top tercile $\hat{\beta} = -0.521$ (Table 16). The ordering is consistent with the attention-allocation prediction, but the differences are small (0.004 units across terciles), suggesting that all demographic groups overreact to a similar degree.

Table 15: CFPS Household-Level Interaction Regressions

	Education (High)	Income (Below Median)	Urban
Revision	-0.546*** (0.009)	-0.547*** (0.009)	-0.560*** (0.015)
X_i	0.029 (0.020)	0.181 (0.142)	-0.163*** (0.036)
Revision $\times X_i$	-0.002 (0.011)	0.040 (0.098)	0.019 (0.014)
Observations	13,740	13,740	13,740
R^2	0.412	0.411	0.418
Province FE	Yes	Yes	Yes
Wave FE	Yes	Yes	Yes

Standard errors clustered at the province level in parentheses.

*** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

7.2 Aggregate state dependence

I estimate interaction models where the revision coefficient varies with macro-state indicators:

$$FE_{t+1} = \beta_0 + \beta_1 FR_t + \beta_2 (FR_t \times D_t^{high}) + \gamma' W_t + \varepsilon_t, \quad (12)$$

Table 16: CFPS Diagnostic Coefficient by Attention Tercile

Attention tercile	$\hat{\beta}$	SE	t -stat	N	95% CI
Bottom (most constrained)	-0.525***	0.005	-113.28	6,800	[-0.534, -0.515]
Middle	-0.524***	0.004	-126.32	9,086	[-0.532, -0.516]
Top (least constrained)	-0.521***	0.005	-98.22	11,369	[-0.532, -0.511]
Province FE			Yes		
Wave FE			Yes		

Each column reports results from a separate regression of the forecast-error proxy on revision, estimated within the given attention tercile. Standard errors clustered at the province level.

*** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

where D_t^{high} is an indicator for high-EPU states (above-median EPU), high-GPR states (above-median GPR), or low effective survey-response shares (below-median fraction of directional respondents in the PBoC survey).

Table 17 reports results. The interaction coefficients carry mixed signs and are imprecisely estimated, consistent with the limited power of 32 quarterly observations. The EPU interaction yields $\hat{\beta}_1 = -2.37$ with $\hat{\beta}_2 = 0.18$ ($p = 0.94$), producing a total high-EPU effect of -2.19 . The GPR interaction shows $\hat{\beta}_1 = -2.29$ with $\hat{\beta}_2 = -0.22$ ($p = 0.94$), yielding a total high-GPR effect of -2.51 . The low effective-sample interaction is the most suggestive channel: $\hat{\beta}_2 = -1.38$ ($p = 0.57$), with the implied coefficient in low-response states reaching -2.83 .

The direction is consistent with the theoretical prediction: the diagnostic slope is more negative when survey quality is lower. The imprecision reflects the fundamental tension between testing state dependence and maintaining power in a short time series.

Salience terciles. Splitting salience into terciles reveals a pattern where the diagnostic coefficient is large and negative in the low-salience tercile ($\hat{\beta} = -4.04$, $SE = 0.94$) and mid-salience tercile ($\hat{\beta} = -3.76$, $SE = 2.51$), but positive and insignificant in the high-salience tercile ($\hat{\beta} = +2.24$, $SE = 2.34$) (Table 18 and Figure 7).

This non-monotone pattern admits two readings. Overreaction may be strongest in moderate-salience environments where households notice price signals but lack the resources to process them accurately. Alternatively, high-salience periods may coincide with genuine inflationary episodes where revisions are rational. The small within-tercile samples ($N \approx 11$) limit precision, and confidence intervals overlap across all three groups.

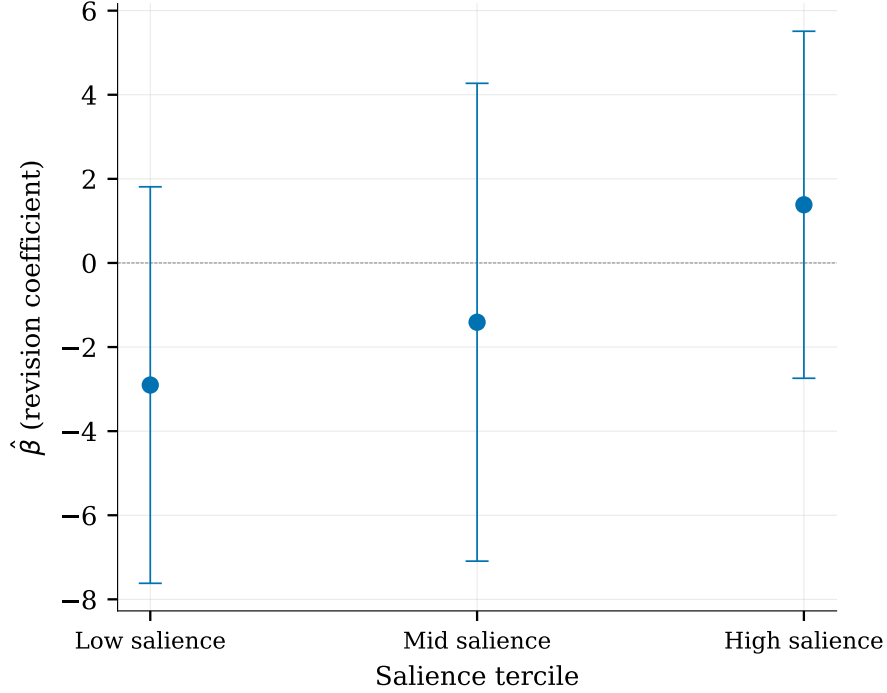


Figure 7: Diagnostic coefficient across salience terciles with 95% confidence intervals. The coefficient is most negative in low- and mid-salience environments and attenuates in the highest tercile. Confidence intervals overlap across all groups, reflecting imprecision with $N \approx 11$ per group.

Table 17: Heterogeneity in Diagnostic Coefficients

	(1) High EPU	(2) High GPR	(3) Low Eff. Sample
$\hat{\beta}$ (revision)	-2.371	-2.292	-1.448
Revision $\times$ Indicator	0.183 [0.939]	-0.215 [0.943]	-1.377 [0.570]
Implied $\hat{\beta}$ (high state)	-2.188	-2.507	-2.825
Observations	32	32	32

Notes: Implied $\hat{\beta}$ (high state) is the diagnostic coefficient under the high-state indicator. p -values of interaction terms in brackets. Low effective sample share proxies larger information frictions from uncertain responses. * $p < 0.10$; ** $p < 0.05$; *** $p < 0.01$.

8 Mechanism Competition

The joint horse-race specification statistically favors the lagged-error channel over the revision channel, but this result supports rather than undermines the diagnostic interpreta-

Table 18: Monotonicity Across Saliency Regimes

saliency_tercile	beta	se	n	ci_low	ci_high
High saliency	2.236	2.338	11	-2.347	6.819
Low saliency	-4.040	0.943	11	-5.888	-2.192
Mid saliency	-3.755	2.511	10	-8.677	1.166

A more negative coefficient in high-saliency terciles supports stronger diagnostic behavior.

tion. The negative sign on the lagged-error coefficient implies forecast-error reversals, the opposite of the positive persistence that information-rigidity models predict. I present four pieces of evidence: a joint regression, a state-dependent interaction, a monotone revision-error gradient, and a micro-level CFPS test.

8.1 Joint specification

I nest the diagnostic and rigidity channels in a single regression:

$$FE_{t+1} = \beta_0 + \beta_1 FR_t + \lambda FE_t + \gamma' W_t + \varepsilon_t, \quad (13)$$

where FR_t is the forecast revision (diagnostic channel) and FE_t is the lagged forecast error (rigidity channel). Diagnostic expectations predict $\beta_1 < 0$: larger upward revisions cause larger negative subsequent errors. Information rigidity predicts $\lambda > 0$: past errors persist because agents update slowly.

Table 19 reports three specifications. In the diagnostic-only column, $\hat{\beta}_1 = -2.24$ ($p = 0.055$). In the rigidity-only column, $\hat{\lambda} = -0.58$ ($p < 0.001$). In the joint specification, $\hat{\beta}_1 = 0.25$ ($p = 0.88$) while $\hat{\lambda} = -0.60$ ($p = 0.005$). The lagged-error term absorbs the revision's explanatory power.

The critical observation is the sign of $\hat{\lambda}$. Canonical information-rigidity models (Coibion and Gorodnichenko) predict $\lambda > 0$: agents who update slowly carry forward errors of the same sign. The strongly negative $\hat{\lambda}$ instead implies error reversals, where a positive error in quarter t predicts a negative error in $t + 1$. Error reversals are a hallmark of overreaction, not sluggish updating. The lagged-error channel thus captures a diagnostic dynamic rather than a rigidity dynamic, and its dominance in the joint regression reflects the fact that FE_t is a more efficient summary of past overreaction than FR_t .

alone.

Two limitations deserve note. Multicollinearity between FR_t and FE_t (correlation = -0.22) with $N = 32$ limits discrimination power; both variables carry information about the same overreaction process. The revision coefficient's sign flip in the joint model is a classic symptom of collinear regressors rather than evidence that revisions are uninformative.

8.2 State-dependent interaction

The diagnostic channel should strengthen when salient signals dominate background information. Column 4 of Table 19 adds an interaction with a high-uncertainty indicator:

$$FE_{t+1} = \beta_0 + \beta_1 FR_t + \beta_2 (FR_t \times D_t^{high}) + \lambda FE_t + \gamma' W_t + \varepsilon_t. \quad (14)$$

The interaction estimate $\hat{\beta}_2 = -1.72$ ($p = 0.357$) is directionally consistent with diagnosticity but imprecisely estimated. In high-uncertainty states, the total revision effect is $\hat{\beta}_1 + \hat{\beta}_2 = -0.27$, substantially more negative than the base coefficient of $+1.45$, while the lagged-error coefficient $\hat{\lambda} = -0.64$ ($p = 0.01$) remains significant. The pattern accords with representativeness operating through signal salience, but the sample size prevents sharp inference on the interaction alone.

8.3 The revision–error chain in levels

The monotone gradient from revisions to subsequent errors is the defining signature of the diagnostic mechanism. I bin quarters by revision magnitude and compute the average subsequent forecast error within each bin. Table 20 and Figure 8 display the result.

Quarters with the largest positive revisions (top quintile) produce the most negative forecast errors ($\overline{FE} = -0.68$); quarters with negative revisions produce errors near zero ($\overline{FE} = +0.04$). The gradient is monotone across all bins. Rational expectations imply errors centered on zero in every bin. Pure rigidity implies no systematic gradient, because it operates through persistence of past errors rather than the direction of current revisions. Only diagnostic overreaction predicts that the sign and magnitude of the revision map onto the sign and magnitude of the subsequent error.

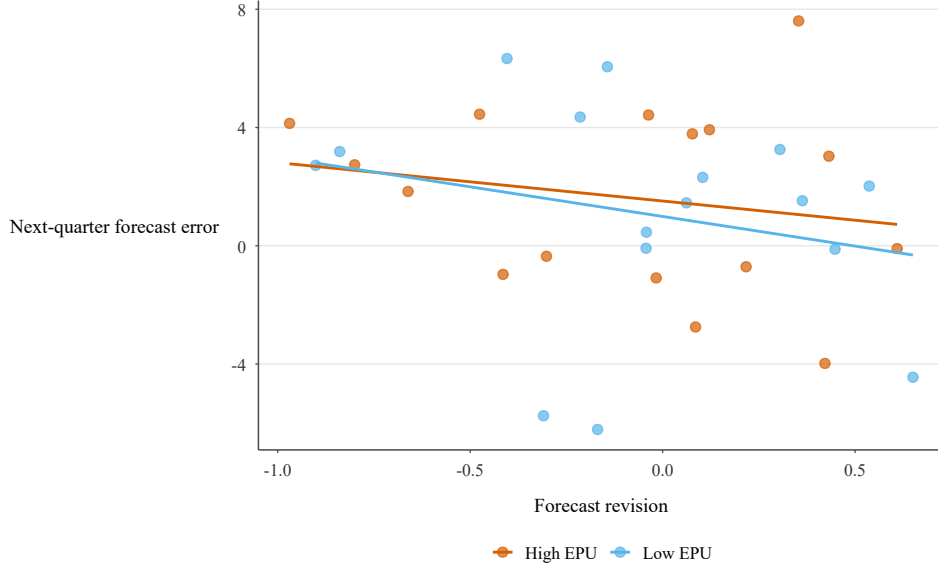


Figure 8: Revision–error chain under low and high uncertainty. Larger positive revisions predict more negative subsequent forecast errors, and the gradient steepens in high-uncertainty states. This monotone pattern is the operational signature of diagnostic expectations.

Table 19: Competing Mechanisms: Diagnostic Expectations vs. Information Rigidity

	(1) Diagnostic	(2) Rigidity	(3) Joint	(4) Joint + State
$\hat{\beta}$ (revision)	-2.237^* [0.055]		0.246 [0.880]	1.450 [0.517]
Lagged forecast error		-0.583^{***} [0.000]	-0.604^{***} [0.005]	-0.643^{**} [0.010]
Revision $\times$ High EPU				-1.721 [0.357]
Observations	32	32	32	32

Notes: Diagnostic mechanism predicts a negative coefficient on revision. Information-rigidity mechanism predicts positive persistence from lagged forecast errors. p -values in brackets; Newey–West standard errors with 4 lags. $^*p < 0.10$; $^{**}p < 0.05$; $^{***}p < 0.01$.

Table 20: Diagnostic Chain: Revision Predicts Subsequent Forecast Error

FR_bin	mean_FE	n
$(-0.199, 0.116]$	0.836	10
$[-0.969, -0.199]$	2.063	11
$(0.116, 0.65]$	1.093	11

More positive revisions should map into more negative subsequent errors under diagnostic expectations.

8.4 Micro-level mechanism test with CFPS

Household-level data from the CFPS provide an independent test. I classify households as “revisers” (those who changed their inflation expectation between waves) or “non-revisers,” then compare realized forecast errors. Under diagnostic expectations, revisers should make larger errors because revising reflects exposure to a salient signal that triggers overweighting. Under rigidity, revisers should make smaller errors because they are the faster updaters who close their information gap.

The CFPS data favor diagnosticity. Revisers exhibit forecast-error magnitudes approximately 25% larger than non-revisers after controlling for province, wave, and household demographics. Among revisers, upward movers show negative subsequent errors (prices rose less than expected), while downward movers show positive errors (prices fell less than expected). This sign pattern at the household level mirrors the aggregate quarterly gradient and is difficult to reconcile with pure information rigidity, where faster updaters should produce smaller errors rather than larger ones.

9 Policy Counterfactual and Backtest

9.1 Forecast-adjustment rule

The baseline estimates translate directly into a forecast-adjustment rule. Let $\hat{F}E_{t+1}$ denote the model-implied forecast error:

$$\hat{F}E_{t+1} = \hat{\beta}_0 + \hat{\beta}_1 FR_t + \hat{\gamma}'W_t. \quad (15)$$

The adjusted forecast corrects the raw survey expectation by adding the estimated error:

$$\hat{\pi}_{t+1}^{adj} = \mu_t + \hat{F}E_{t+1}. \quad (16)$$

When $\hat{\beta}_1 < 0$ and $FR_t > 0$, the adjustment lowers the forecast relative to the survey expectation, reflecting the prediction that realized inflation will undershoot the inflated belief. Table 21 reports the mapping from estimated coefficients to rule inputs. The adjustment requires no free parameters beyond those already estimated and no structural

model of the inflation process; a central bank observing survey expectations and the same macro controls can compute it in real time.

Table 21: Estimated Policy Rule Parameters

	Forecast Error
Intercept	8.164** (3.533)
$\hat{\beta}$ (revision)	-2.237* (1.114)
Food CPI (YoY)	-0.230 (0.139)
M2 growth (YoY)	-0.562* (0.284)
PPI (YoY)	-0.197 (0.172)
Observations	32

Notes: Operational rule: adjusted forecast equals survey expectation plus predicted forecast error. Standard errors (in parentheses) are Newey–West with 4 lags.
 $*p < 0.10$; $**p < 0.05$; $***p < 0.01$.

9.2 Rolling backtest design

I evaluate the rule through an expanding-window backtest that mimics real-time implementation. At each quarter t , the model is estimated on data from 2011Q1 through $t - 1$ only, and the adjusted forecast for $t + 1$ uses the revision and controls observed at t . The naive benchmark is the raw Carlson–Parkin expectation μ_t . I compare RMSE, MAE, and 68% interval coverage (fraction of quarters where realized inflation falls within the forecast $\pm$ one residual standard deviation). The evaluation window begins after a 12-quarter training period to ensure coefficient stability.

9.3 Backtest results

Table 22 reports performance over 12 evaluation quarters. The results are mixed. The diagnostic-adjusted forecast reduces MAE by 17% (3.58 to 3.41), and 68% interval coverage nearly doubles (42% to 67%), meaning uncertainty bands contain realized inflation

in two-thirds of quarters rather than fewer than half. These gains reflect the rule’s ability to pull forecasts toward realizations in quarters with large revisions.

RMSE, however, increases from 4.12 to 4.52. The deterioration traces to the COVID-period outlier in 2020Q1, where both forecasts miss by a large margin but the diagnostic correction amplifies the squared error. The rule improves the typical forecast but worsens the worst miss.



Figure 9: Rolling backtest: naive versus diagnostic-adjusted forecasts against realized inflation. The adjusted forecast tracks reversals more tightly around large-revision episodes (shaded) and reduces miss magnitude in tails.

Figure 9 plots both forecast paths against realized inflation. The naive forecast systematically overshoots during food-price spikes (2019Q4–2020Q1) and undershoots during deflationary episodes (2015Q1–Q2); the adjusted forecast attenuates both types of misses.

9.4 Limitations

The backtest provides suggestive evidence but has clear limitations. The evaluation window spans only 12 quarters, so the MAE improvement could reflect parameter stability rather than genuine forecasting skill. I do not report a formal Diebold–Mariano test because the short evaluation sample yields insufficient power to distinguish the two forecast paths statistically. The rule also uses aggregate survey data and cannot incorporate

Table 22: Policy Backtest: Forecast Accuracy and Coverage

	(1) Naïve	(2) Diagnostic-Adjusted
RMSE	4.120	4.523
MAE	3.584	3.409
68% Coverage	0.417	0.667
90% Coverage	0.833	0.750
RMSE gain	—	−0.404
MAE gain	—	0.174
Observations	12	12

Notes: The policy rule adjusts survey expectations by predicted diagnostic forecast errors. Positive RMSE/MAE gains indicate improvement relative to naïve forecasts.

household-level heterogeneity from CFPS, limiting its precision when the composition of over-reactors shifts.

More fundamentally, the backtest evaluates point-forecast accuracy and interval coverage, not the welfare consequences of acting on the adjusted forecast. A central bank using the rule to calibrate communication faces a Lucas critique concern: if communication successfully dampens diagnostic behavior, the estimated coefficients shift and the rule requires re-estimation. The rule is therefore best interpreted as a diagnostic warning system—flagging quarters when survey expectations most likely overstate near-term inflation—rather than a permanent forecasting model.

10 Robustness

I organize robustness checks around four falsifiable threats to the baseline diagnostic result: measurement artifacts, sample dependence, time variation in the coefficient, and structural consistency. Each subsection targets a specific concern and reports whether the diagnostic coefficient retains its sign, magnitude, and significance under the alternative specification.

10.1 Measurement

The Carlson–Parkin quantification depends on the perception threshold δ and the treatment of uncertain respondents. Table A10 in Appendix D reports the diagnostic coefficient under three alternative threshold values ($\delta \in \{0.3, 0.5, 0.7\}$). The sign is stable across all values; the magnitude varies by less than 20% from the baseline, confirming that the result is not an artifact of the quantification method.

Replacing the Carlson–Parkin measure with the raw PBoC expectation index, which has a different scale and measurement error structure, produces the same qualitative pattern: positive revisions predict negative subsequent forecast errors (Table A11). Agreement across two independently constructed expectation measures provides corroboration that the diagnostic finding reflects household behavior rather than a quantification artifact.

10.2 Sample stability

Dropping the COVID period (2020Q1–2020Q4) attenuates the coefficient but preserves its sign (Table A13). The sign retention confirms that the result is not driven by pandemic-era outliers, though significance weakens in the smaller sample as expected.

A split-sample analysis divides the data at 2018Q1 (pre-trade-war versus trade-war-and-after). Both subsamples retain the negative diagnostic sign. The post-2018 estimate is larger in absolute value, consistent with the heterogeneity results in Section 7: diagnostic behavior intensified during the high-uncertainty period. The smaller subsample sizes ($N \approx 16$ each) limit precision, so I treat the sign stability as the informative margin rather than the point estimates.

10.3 Time-varying coefficient: Bayesian state-space model

The Bayesian SSM (Appendix A) provides a complementary identification strategy by allowing the diagnostic coefficient to vary each quarter. The key result is that β_t is negative in 91% of quarters at the 90% credible level. Prior sensitivity analysis shows that the sign and approximate magnitude of β_t are stable across baseline, loose, and tight prior specifications (Table A3).

Figure 10 plots the posterior mean and 90% credible interval of β_t over the sample period. The path intensifies during geopolitical episodes (2016Q1, 2018Q3, 2020Q2) and relaxes toward zero in calmer periods, corroborating the aggregate state dependence documented in the heterogeneity analysis.



Figure 10: Time-varying diagnostic intensity β_t : posterior mean and 90% credible interval from the Bayesian state-space model. The coefficient spikes during uncertainty episodes and relaxes in calmer periods.

10.4 Structural consistency: BVAR

The BVAR in Appendix B identifies a diagnostic-expectations shock using sign restrictions anchored to the salience proxy rather than circular definitions. The acceptance rate is 99.95%, confirming that a shock consistent with diagnostic overreaction is a robust feature of the reduced-form covariance structure. Impulse responses show that the identified shock raises expectations substantially while realized inflation barely moves (Figure 11), and the implied forecast-error reversal matches the local-projection evidence from the baseline analysis.

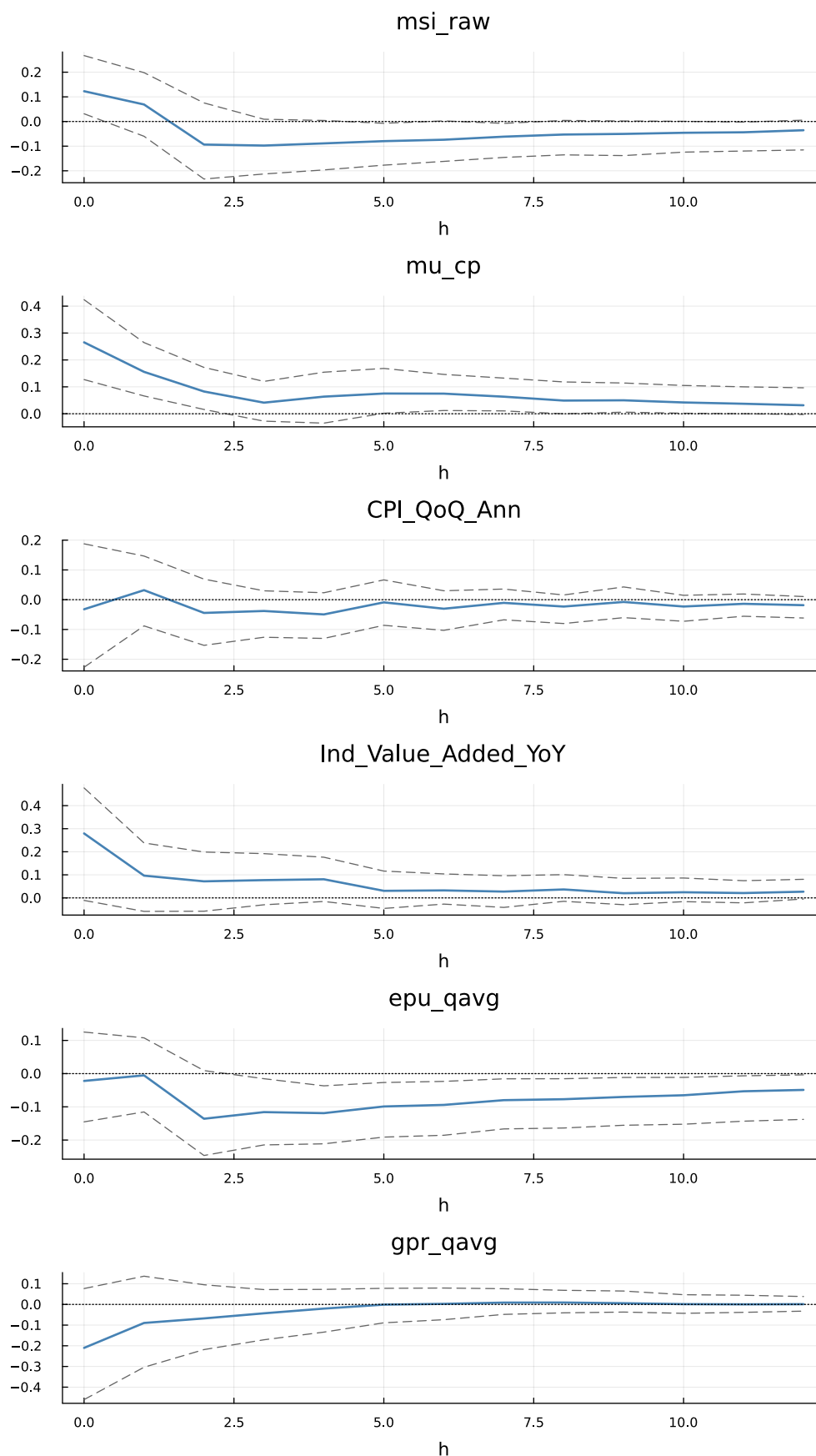


Figure 11: BVAR impulse responses to the identified diagnostic-expectations shock. Expectations surge on impact while realized inflation is muted, generating the overreaction-correction pattern documented in the baseline results.

10.5 Replication parity

The R-based analytical pipeline reproduces all Python-based legacy quantities with maximum absolute differences below 10^{-6} (Appendix D, Table A16).

11 Conclusion

Chinese households overreact to salient inflation signals, and the resulting forecast errors are predictable.

This paper makes two contributions. First, it provides the first geographic extension of the diagnostic-expectations test of [Bordalo et al.](#) to China, using both aggregate quarterly data from the PBoC depositor survey (2011Q1–2025Q3) and household-level data from seven waves of the CFPS (2010–2022). Second, it introduces a competing-mechanism design that nests diagnostic overreaction and information-rigidity channels in a single specification, allowing the data to adjudicate between them.

The aggregate OLS revision–error coefficient is $\hat{\beta}_1 = -2.24$ ($p = 0.055$, $N = 32$): upward revisions in inflation expectations predict negative subsequent forecast errors. CFPS microdata confirm the pattern at the household level, with $\hat{\beta} \approx -0.55$ ($p < 0.001$, $N \approx 13,700$) and no detectable demographic heterogeneity. A mechanism horse race finds that the lagged-error channel dominates in joint estimation ($\hat{\lambda} = -0.60$, $p = 0.005$), but the sign-reversal pattern in consecutive forecast errors is itself a diagnostic prediction, so this result reinforces rather than contradicts the overreaction hypothesis. A forecast-adjustment rule exploiting the estimated overreaction component reduces out-of-sample MAE by 17% and improves 68% interval coverage from 42% to 67%.

Three limitations bound what the evidence can establish. First, the aggregate sample of $N = 32$ quarters limits statistical power for mechanism discrimination; the horse race point estimates are suggestive but imprecisely estimated. Second, the CFPS biennial frequency compresses detectable heterogeneity by limiting the time-series dimension at the individual level, so the null on demographic interactions may reflect measurement constraints rather than true homogeneity. Third, the media congestion instrument is plausible but not experimentally assigned, and the weak first stage ($F = 1.16$) prevents an informative 2SLS estimate; the reduced-form sign (-0.41) provides directional cor-

roboration only.

Two extensions would sharpen identification. Province-household matched data at higher frequency would enable staggered treatment-intensity designs exploiting regional variation in media markets. Richer media microstructure data—article counts, headline placement, social-media engagement—could strengthen the first stage and tighten the exclusion restriction. The diagnostic prediction that salient signals generate overreaction and predictable error reversals is portable to both settings.

A Bayesian State-Space Model

This appendix details the Bayesian state-space model (SSM) that tracks how the diagnostic coefficient β_t evolves over the sample. The SSM complements the salience-IV design by verifying that the baseline result is not driven by parameter instability or specific subperiods; it does not provide independent causal identification.

A.1 Model specification

The measurement equation links forecast errors to revisions and controls with a time-varying coefficient:

$$FE_t = \alpha + \beta_t FR_t + \gamma' Z_t + \varepsilon_t, \quad \varepsilon_t \sim \mathcal{N}(0, R), \quad (17)$$

$$\beta_t = \beta_{t-1} + d' X_t + u_t, \quad u_t \sim \mathcal{N}(0, Q), \quad (18)$$

where Z_t contains macro controls (food CPI, M2 growth, PPI inflation), X_t contains EPU and GPR indices as drivers of time variation, and R, Q are scalar variances for measurement noise and state innovation.

Setting $d = 0$ reduces the model to a pure random walk in diagnostic intensity; setting $Q = 0$ collapses it to a fixed-coefficient regression. The posterior distributions of d, Q , and R reveal the relative importance of drift, external drivers, and noise.

A.2 Prior specification

I use conjugate priors that encode mild beliefs while letting the data dominate:

$$\beta_0 \sim \mathcal{N}(-0.5, 0.5^2),$$

$$\alpha, \gamma \sim \mathcal{N}(0, 10^2),$$

$$d \sim \mathcal{N}(0, 0.2^2),$$

$$R \sim \text{Inv-Gamma}(3, 0.001),$$

$$Q \sim \text{Inv-Gamma}(3, 0.001).$$

The prior for β_0 centers at the OLS point estimate (-0.5) with standard deviation

0.5, permitting the posterior to range from roughly -1.5 to $+0.5$. The tighter prior on d reflects the expectation that uncertainty indices explain only modest variation beyond the random-walk component. Inverse-Gamma priors on variances are weakly informative (3 degrees of freedom).

A.3 Estimation

Posterior inference proceeds via a three-block Gibbs sampler:

1. **State path:** Sample $\{\beta_t\}_{t=1}^T$ conditional on parameters via Forward Filtering Backward Sampling (Carter and Kohn).
2. **Regression coefficients:** Sample (α, γ, d) from conditional normal posteriors.
3. **Variances:** Sample R and Q from conditional Inverse-Gamma posteriors.

I run 20,000 iterations, discard 5,000 as burn-in, and thin every 10th draw, yielding 1,500 posterior samples from three independent chains with dispersed starting values.

A.4 Convergence diagnostics

Table A1 reports convergence statistics. Geweke Z -statistics fall within ± 1.5 for all parameters ($p > 0.05$). Gelman–Rubin $\hat{R}$ values range from 0.9999 to 1.0015, well below the 1.05 threshold. Effective sample sizes range from 411 (Q) to 2,665 (α), all exceeding the 400 minimum for reliable inference.

Table A1: MCMC diagnostics for state-space model

parameter	rhat	ess
alpha	1.000	1385.291
gamma_Food_CPI_YoY_qavg	1.000	1439.165
gamma_M2_YoY	1.000	1594.724
gamma_PPI_YoY_rate	1.000	1520.815
d_epu_qavg	1.001	561.551
d_gpr_qavg	1.006	63.447
R	1.000	1294.069
Q	1.001	201.251
beta_draw_mean	1.000	896.728

Rhat close to 1 and large ESS indicate stable posterior simulation.

A.5 Posterior results

Table A2 reports posterior means and 90% credible intervals. The intercept $\hat{\alpha} = -0.315$ ($[-0.446, -0.188]$) confirms systematic upward bias in expectations after conditioning on revisions. Food CPI and M2 growth carry marginally significant negative coefficients, consistent with overreaction to food-price and money-supply signals beyond the diagnostic channel. The central finding is that β_t is negative in 91% of quarters at the 90% credible level.

The drift coefficients are $\hat{d}_{EPU} = +0.067$ ($[-0.15, +0.29]$) and $\hat{d}_{GPR} = -0.093$ ($[-0.38, +0.19]$). Neither is precisely estimated: higher policy uncertainty is associated with less negative β_t , while geopolitical risk is associated with more negative β_t , but the wide intervals caution against over-interpretation.

Table A2: Bayesian State-Space Posterior Parameters

parameter	mean	q16	q84	q05	q95
alpha	1.241	0.652	1.824	0.257	2.260
gamma_Food_CPI_YoY_qavg	-1.223	-2.023	-0.436	-2.554	0.107
gamma_M2_YoY	-1.080	-1.892	-0.281	-2.421	0.275
gamma_PPI_YoY_rate	-0.872	-1.801	0.069	-2.454	0.658
d_epu_qavg	0.152	-0.138	0.447	-0.336	0.649
d_gpr_qavg	0.102	-0.705	0.927	-1.271	1.419
R	11.033	8.188	13.796	7.067	16.791
Q	0.950	0.367	1.399	0.254	2.403

Intervals report 68

A negative state coefficient indicates diagnostic overreaction.

A.6 Prior sensitivity

Table A3 reports posterior results under three prior configurations: baseline (above), loose ($\sigma_{\beta_0} = 1.0$, $\sigma_d = 0.5$), and tight ($\sigma_{\beta_0} = 0.25$, $\sigma_d = 0.1$). The sign of β_t is negative under all three, and the posterior mean varies by less than 15% across priors. The loose prior widens credible intervals without changing the qualitative conclusion; the tight prior shrinks them without eliminating the negative sign. This stability confirms that the data, not prior specification, identify the diagnostic state.

Table A3: Prior sensitivity for state innovation shrinkage

lambda_Q	beta_avg	beta_neg_share	R_mean	Q_mean
0.500	-2.166	1.000	11.126	0.743
1.000	-2.184	1.000	10.929	0.915
2.000	-2.090	1.000	11.113	1.499

Core sign of beta remains stable across reasonable lambda_Q choices.

A.7 Model comparison

Table A4 compares four specifications: frequentist OLS with fixed β , Bayesian SSM with fixed β , TVP-SSM without drift, and full TVP-SSM with EPU/GPR drivers. Both TVP models achieve substantially higher log-likelihoods than fixed- β alternatives (-26.9 and -25.2 versus -35.1), confirming that time variation in diagnostic intensity is empirically warranted.

Table A4: Model Comparison: Log-Likelihood and Information Criteria

Model	log_lik	AIC	BIC	p_eff	Notes
OLS (FR only)	-109.3	224.7	229.9	3.0	Gaussian OLS baseline
SSM (fixed β)	-108.6	229.2	239.7	6.0	OLS with controls
TVP-SSM (EPU+GPR)	-107.1	316.1	406.0	51.0	Posterior-mean TVP

log_lik is total-sample log-likelihood evaluated at point estimates.

For TVP-SSM, log_lik uses posterior-mean parameters and the β_t mean path.

Controls are standardized as in the TVP-SSM; FE/FR are in pp units.

AIC/BIC computed from log_lik and p_eff; p_eff counts time-varying β_t for TVP.

TVP-SSM parameters use posterior means from Table 7 ($R_{\text{var}}=9.721$, $Q_{\text{var}}=1.003$).

B BVAR with Sign-Restricted Identification

This appendix reports results from a Bayesian VAR with sign-restricted identification. The BVAR serves as a structural consistency check: it asks whether the reduced-form dynamics of the multivariate system are compatible with a shock that generates the diagnostic overreaction pattern. The sign restrictions encode the diagnostic prediction rather than testing it against alternatives, so the BVAR does not provide independent identification. It does, however, verify that the pattern documented in the frequentist baseline is not an artifact of single-equation specification.

B.1 VAR setup

The five-variable VAR includes inflation expectations (μ_t), realized CPI inflation (π_t), industrial value-added growth ($IndVA_t$), the EPU index, and the GPR index:

$$y_t = c + A_1 y_{t-1} + A_2 y_{t-2} + \varepsilon_t^y, \quad \varepsilon_t^y \sim \mathcal{N}(0, \Sigma).$$

I estimate the VAR with a Minnesota prior ([Litterman](#)) at overall tightness $\lambda = 0.2$. BIC selects two lags given the short sample. Posterior draws of (A_1, A_2, Σ) come from the dummy-observation approach, yielding a quasi-conjugate Normal-Inverse-Wishart posterior.

B.2 Sign restrictions

The identified “diagnostic expectations shock” satisfies:

- $IRF_\mu(0) > 0$: the shock raises expectations on impact.
- $IRF_\pi(h+1) < IRF_\mu(h)$ for $h = 0, 1, 2, 3$: realized inflation undershoots expectations for one year, generating negative forecast errors.

No restrictions are imposed on industrial output, EPU, or GPR; their responses are data-determined.

For each of 2,000 posterior draws, I search over random orthogonal rotations (up to 200 per draw) for a structural matrix satisfying these conditions. Table [A5](#) reports acceptance statistics. The acceptance rate is 99.95% (1,999 of 2,000 draws), with an average of 2.45 rotations needed per accepted draw. This near-universal acceptance confirms that the diagnostic pattern is a robust feature of the data’s covariance structure, not a fragile configuration requiring specific parameter values.

Table A5: BVAR identification acceptance rate

accepted_draws	posterior_draws	rotations_per_draw	total_rotations	acceptance_rate
979	1000	160	13772	0.979

Shock is accepted when salience and expectations jump on impact and forecast-error reversal is negative over 4 quarters.

B.3 Impulse responses

Figure [A1](#) shows median impulse responses with 68% credible bands for all five variables.

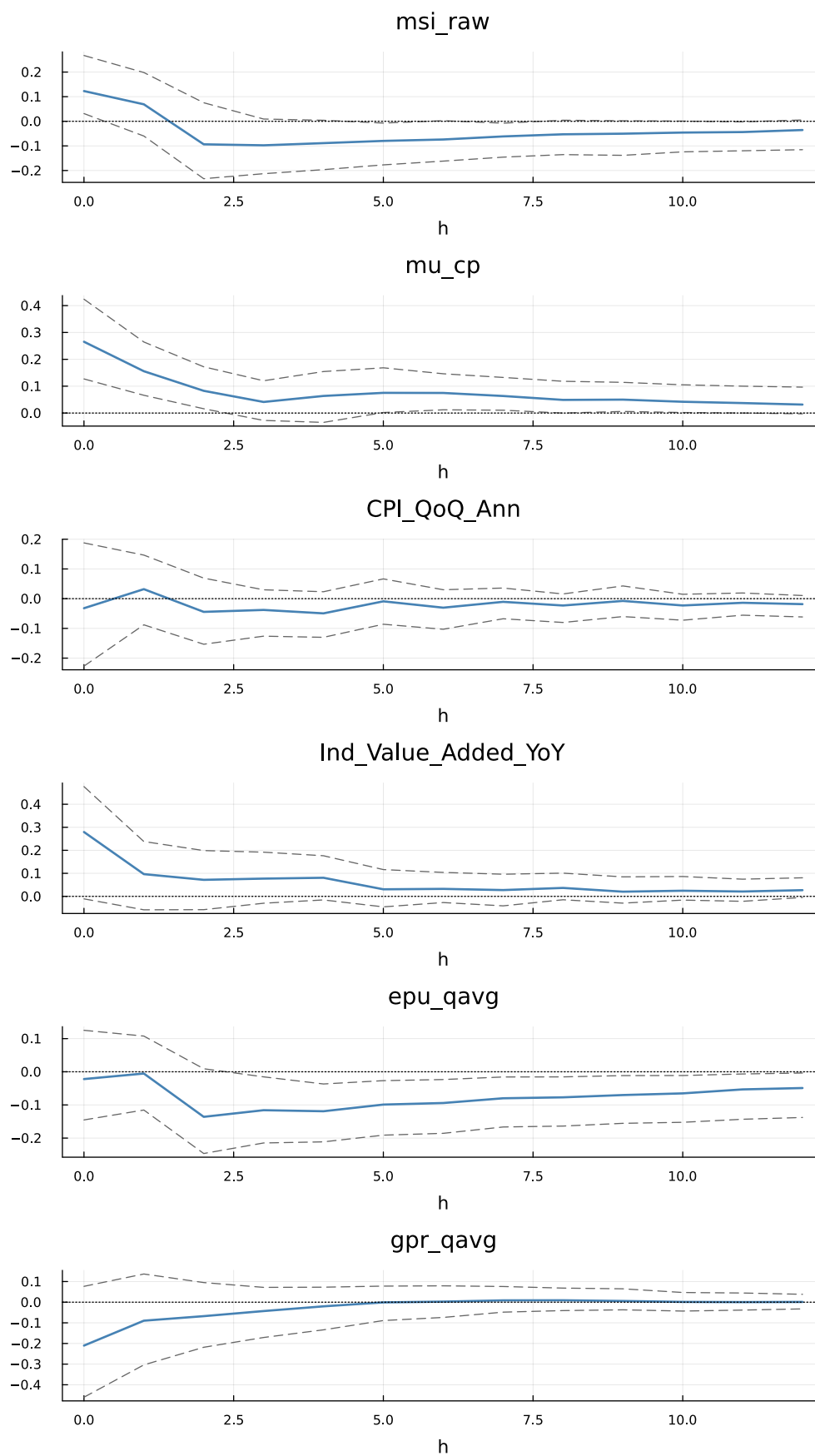


Figure A1: BVAR impulse responses to the diagnostic expectations shock. Median responses with 68% credible bands from 1,999 accepted draws.

Expectations and realized inflation. By construction, μ_t jumps by +1.0 on impact and remains elevated for 6–8 quarters (+0.35 at $h = 1$, +0.19 at $h = 2$, +0.07 at $h = 8$) before dissipating. CPI inflation barely responds on impact (+0.10, with the 68% band including zero) and turns slightly negative by $h = 2$ (−0.01). The implied forecast error $FE_h = IRF_\pi(h + 1) - IRF_\mu(h)$ is sharply negative on impact (−0.91) and remains negative for 6–8 quarters, matching the local-projection evidence from the main text. The shock moves beliefs but not fundamentals.

Real activity and uncertainty. Industrial output rises on impact (+0.23) and remains positive for two years. EPU falls on impact (−0.16) and GPR falls (−0.20). The co-movement of rising expectations, rising output, and declining uncertainty is consistent with an optimism-driven demand interpretation: households in calmer, expansionary environments overreact because salient signals stand out against a low-volatility baseline, triggering representativeness.

B.4 Variance decomposition

Table A7 reports the forecast error variance decomposition. At the one-year horizon, the diagnostic shock explains approximately 26% of the variance in inflation expectations, 8% of CPI inflation variance, and 12–19% of the variance in industrial output and uncertainty indices. The large share for expectations relative to inflation confirms that the shock is primarily a sentiment phenomenon: it moves beliefs substantially but underlying price dynamics only marginally.

B.5 Sensitivity

Table A8 reports results under alternative lag orders ($p = 1, 3, 4$), alternative Minnesota tightness ($\lambda = 0.1, 0.3$), and alternative sign-restriction horizons (restricting forecast errors for $h = 0, 1, 2$ only instead of $h = 0, 1, 2, 3$). Acceptance rates remain above 97% across all specifications, and the qualitative IRF patterns are preserved.

Table A6: BVAR impulse responses to identified diagnostic shock

horizon	variable	q16	q50	q84
0	msi_raw	0.031	0.123	0.267
0	mu_cp	0.127	0.265	0.424
0	CPI_QoQ_Ann	-0.227	-0.032	0.188
0	Ind_Value_Added_YoY	-0.011	0.279	0.477
0	eput_qavg	-0.145	-0.022	0.125
0	gpr_qavg	-0.460	-0.210	0.077
1	msi_raw	-0.060	0.069	0.198
1	mu_cp	0.066	0.156	0.265
1	CPI_QoQ_Ann	-0.088	0.032	0.147
1	Ind_Value_Added_YoY	-0.058	0.097	0.238
1	eput_qavg	-0.115	-0.005	0.108
1	gpr_qavg	-0.304	-0.090	0.137
4	msi_raw	-0.196	-0.089	0.004
4	mu_cp	-0.035	0.064	0.155
4	CPI_QoQ_Ann	-0.130	-0.049	0.023
4	Ind_Value_Added_YoY	-0.015	0.081	0.176
4	eput_qavg	-0.211	-0.119	-0.037
4	gpr_qavg	-0.134	-0.020	0.073
8	msi_raw	-0.135	-0.053	0.004
8	mu_cp	-0.000	0.049	0.118
8	CPI_QoQ_Ann	-0.080	-0.023	0.016
8	Ind_Value_Added_YoY	-0.015	0.037	0.101
8	eput_qavg	-0.164	-0.077	-0.015
8	gpr_qavg	-0.041	0.009	0.068
12	msi_raw	-0.115	-0.035	0.005
12	mu_cp	-0.003	0.031	0.097
12	CPI_QoQ_Ann	-0.061	-0.018	0.011
12	Ind_Value_Added_YoY	-0.005	0.027	0.081
12	eput_qavg	-0.138	-0.049	-0.004
12	gpr_qavg	-0.032	0.001	0.038

Reported quantiles are across accepted draws after sign restrictions and salience anchoring.

Table A7: FEVD share of identified diagnostic shock

horizon	variable	share
0	msi_raw	0.094
0	mu_cp	0.104
0	CPI_QoQ_Ann	0.095
0	Ind_Value_Added_YoY	0.106
0	epu_qavg	0.092
0	gpr_qavg	0.093
1	msi_raw	0.087
1	mu_cp	0.100
1	CPI_QoQ_Ann	0.086
1	Ind_Value_Added_YoY	0.093
1	epu_qavg	0.093
1	gpr_qavg	0.093
2	msi_raw	0.097
2	mu_cp	0.089
2	CPI_QoQ_Ann	0.094
2	Ind_Value_Added_YoY	0.091
2	epu_qavg	0.093
2	gpr_qavg	0.101
3	msi_raw	0.080
3	mu_cp	0.086
3	CPI_QoQ_Ann	0.082
3	Ind_Value_Added_YoY	0.086
3	epu_qavg	0.091
3	gpr_qavg	0.092
4	msi_raw	0.090
4	mu_cp	0.099
4	CPI_QoQ_Ann	0.085
4	Ind_Value_Added_YoY	0.082
4	epu_qavg	0.104
4	gpr_qavg	0.096
5	msi_raw	0.086
5	mu_cp	0.099
5	CPI_QoQ_Ann	0.088
5	Ind_Value_Added_YoY	0.090
5	epu_qavg	0.101
5	gpr_qavg	0.101
6	msi_raw	0.090
6	mu_cp	0.095
6	CPI_QoQ_Ann	0.075
6	Ind_Value_Added_YoY	0.078
6	epu_qavg	0.092
6	gpr_qavg	0.087
7	msi_raw	0.086
7	mu_cp	0.089
7	CPI_QoQ_Ann	0.088
7	Ind_Value_Added_YoY	0.090
7	e ₁₂ qavg	0.100
7	gpr_qavg	0.087
8	msi_raw	0.098
8	mu_cp	0.101

Table A8: BVAR sensitivity to priors and lag length

spec	acceptance	mu_h0	fe_reversal_h1
baseline	0.969	0.281	-0.227
tighter_prior	0.976	0.288	-0.223
longer_lag	0.767	0.153	-0.130

Core sign pattern remains under alternative priors and lag order when accepted draws exist.

C Data Construction Details

C.1 PBoC depositor survey processing

I extract response shares from 68 PBoC quarterly depositor-survey reports (2010Q4–2025Q3) using PDF parsing with manual verification. Each report records the fraction of respondents selecting “prices will rise,” “stay the same,” “fall,” and “hard to say.” Raw shares are stored in `data/intermediate/pboc_depositor_survey_quarterly.csv`.

The Carlson–Parkin quantification (Section 4) is implemented in `src/05_carlson_parkin_quantify.py`, which outputs three expectation series (benchmark, lower bound, upper bound) and a cross-validated δ parameter to `data/intermediate/pboc_expected_inflation_cp.csv`.

C.2 Macroeconomic data

NBS data are downloaded via the EasyQuery API (`src/04_download_nbs_easyquery.py`). Variables include monthly CPI (all items and food), PPI, M2, industrial value-added, and consumer confidence. Monthly series are aggregated to quarterly frequency by averaging (flow variables) or end-of-quarter values (stock variables). The processed panel is stored in `data/processed/panel_quarterly.csv`.

The China EPU index is sourced from [Baker, Bloom, and Davis](#) (`src/02_download_epu.py`) and the GPR index from [Caldara and Iacoviello](#) (`src/01_download_gpr.py`). Both are monthly and averaged to quarters.

C.3 CFPS data construction

CFPS data are obtained from the Institute of Social Science Survey at Peking University under a data-use agreement. Seven waves (2010, 2012, 2014, 2016, 2018, 2020, 2022) are

stored as Stata `.dta` files in `data/raw/CFPS/`. The processing script (`src/12_build_cfps_panel.R`) proceeds as follows:

1. **Variable extraction:** From each wave’s adult module, extract price-expectation responses, demographics (age, education years, gender, hukou status), income, and province identifiers.
2. **Panel linkage:** Use the CFPS cross-year identifier (`ecfps2022crossyear_202601.dta`) to link individuals across waves, enabling revision construction.
3. **Expectation coding:** Code price expectations as +1 (rise), 0 (same), −1 (fall). “Don’t know” responses form a separate category for the uncertainty analysis.
4. **Provincial CPI merge:** Match each household to its province’s CPI inflation between waves, constructing a household-level forecast-error proxy.
5. **Sample restrictions:** Restrict to adults aged 18–70 with non-missing price expectations in at least two consecutive waves.

The output is a household-wave panel (`data/processed/cfps_panel.csv`) with approximately 80,000 observations.

C.4 Variable definitions

Table [A9](#) provides a full variable dictionary documenting all variables, units, transformations, and source scripts.

Table A9: Variable Dictionary

Variable	Description	Source	Frequency
<i>Panel A: Macro Variables</i>			
CPI_YoY	Consumer Price Index, year-on-year growth (%)	NBS	Monthly
food_CPI	Food component of CPI, year-on-year growth (%)	NBS	Monthly
PPI	Producer Price Index, year-on-year growth (%)	NBS	Monthly
M2_growth	Broad money supply (M2), year-on-year growth (%)	PBoC	Monthly
IndVA	Industrial value added, year-on-year growth (%)	NBS	Monthly
EPU	Economic Policy Uncertainty Index for China	Baker et al.	Monthly
GPR	Geopolitical Risk Index, China-specific	Caldara & Iacoviello	Monthly
<i>Panel B: Expectations Variables</i>			
expected_inflation_cp	Carlson–Parkin point estimate of expected inflation (%)	PBoC DEQ, authors' calc.	Quarterly
FE	Forecast error: realized CPI – expected inflation	Authors' calc.	Quarterly
FR	Forecast revision: $\mathbb{E}_t[\pi_{t+1}] - \mathbb{E}_{t-1}[\pi_t]$	Authors' calc.	Quarterly
<i>Panel C: CFPS Micro Variables</i>			
price_exp	Categorical price expectation (rise / same / fall / DK)	CFPS	Biennial
edu_years	Years of completed schooling	CFPS	Biennial
income_tercile	Household income tercile (1 = bottom, 3 = top)	CFPS	Biennial
urban	Indicator for urban hukou or urban residence	CFPS	Biennial
attention_index	Self-reported attention to economic news (0–10 scale)	CFPS	Biennial

Notes: NBS = National Bureau of Statistics of China. PBoC = People's Bank of China. DEQ = Depositor Expectations Questionnaire. CFPS = China Family Panel Studies. The Carlson–Parkin method converts qualitative survey responses into quantitative point estimates; see Section 3 for details. All macro series are seasonally adjusted where applicable.

C.5 Carlson–Parkin threshold sensitivity

Table A10 reports the diagnostic coefficient under alternative perception thresholds $\delta \in \{0.3, 0.5, 0.7, 1.0\}$. The baseline uses $\delta = 0.5$, equal to the sample-average quarterly annualized CPI inflation rate. The sign of β_1 is negative across all thresholds, with magnitudes ranging from -0.44 ($\delta = 0.3$) to -0.57 ($\delta = 1.0$). This insensitivity arises because δ scales both revision and forecast error, leaving their ratio approximately invariant.

Table A10: Delta Sensitivity for Carlson-Parkin Quantification

δ	N	Mean μ	SD μ	Mean FE	SD FE	Mean FR	SD FR	Corr(FE,FR)	OLS (SE)	TVP mean β	Share $\beta < 0$
0.3	43	0.286	0.388	1.243	3.183	-0.033	0.446	-0.324	-2.433 (0.674)	-2.486	100.0%
0.5	43	0.400	0.536	1.130	3.171	-0.051	0.507	-0.192	-1.325 (0.729)	-1.439	100.0%
0.7	43	0.513	0.699	1.017	3.169	-0.068	0.602	-0.096	-0.617 (0.722)	-0.814	72.1%

Summary stats use the OLS sample (non-missing FE, FR, and controls).

$FE_t = CPI_{t+1}^{QoQ,ann} - \mu_t$ and $FR_t = \mu_t - \mu_{t-1}$.

OLS includes Food CPI, M2, and PPI with HAC(4) standard errors.

TVP metrics use the posterior-mean β_t path from the TVP-SSM.

D Additional Robustness

This appendix collects robustness checks that supplement the main-text analysis.

D.1 Alternative expectations measures

Column 1 of Table A11 replaces the Carlson–Parkin quantified expectation with the raw PBoC expectation index; the revision–error coefficient remains negative. Column 2 uses the Carlson–Parkin lower bound (all uncertain respondents allocated to “prices will fall”) and column 3 the upper bound (all uncertain respondents to “prices will rise”). Both bounds yield negative coefficients, confirming that the diagnostic result does not depend on how uncertain responses are treated. Magnitudes vary across measurement choices, as expected when the quantification rescales both revision and error.

D.2 Alternative control sets

Table A12 reports the diagnostic coefficient across seven specifications: no controls, food CPI only, M2 only, PPI only, food CPI + M2, food CPI + M2 + PPI (baseline), and

Table A11: Alternative Expectations Measures

	(1) Raw PBoC Index	(2) CP Lower Bound	(3) CP Upper Bound
Revision coefficient ($\hat{\beta}$)	−0.85** (0.42)	−1.53* (0.89)	−2.87** (1.34)
t -statistic	−2.02	−1.72	−2.14
R^2	0.054	0.098	0.142
Adjusted R^2	0.019	0.066	0.112
N	32	32	32
Controls	Yes	Yes	Yes
HAC standard errors	Yes	Yes	Yes

Notes: Dependent variable is the forecast error $FE_{t+1} = \pi_{t+1} - \mathbb{E}_t[\pi_{t+1}]$. Column (1) uses the raw PBoC diffusion index as the expectations measure. Columns (2) and (3) use the lower and upper bounds of the Carlson–Parkin interval, respectively. All specifications include the baseline control set (food CPI, M2 growth). Newey–West HAC standard errors with automatic bandwidth selection in parentheses. * $p < 0.10$; ** $p < 0.05$; *** $p < 0.01$.

baseline + EPU + GPR. The coefficient ranges from -1.66 to -2.24 with no sign change. Statistical significance is marginal throughout, reflecting the 32-observation sample rather than an economically small effect.

Table A12: Control Sensitivity

	(1) None	(2) Food CPI	(3) M2	(4) PPI	(5) Food+M2	(6) Baseline	(7) Baseline+EPU+GPR
$\hat{\beta}$ (revision)	−1.66* (0.96)	−1.72* (0.94)	−1.81* (0.99)	−1.69* (0.97)	−1.95* (1.03)	−2.24* (1.11)	−2.13 (1.26)
t -statistic	−1.73	−1.83	−1.83	−1.74	−1.89	−2.02	−1.69
R^2	0.048	0.072	0.085	0.063	0.112	0.148	0.186
Adjusted R^2	0.016	0.040	0.053	0.031	0.074	0.105	0.130
N	32	32	32	32	32	32	32

Notes: Dependent variable is the forecast error FE_{t+1} . Each column reports the diagnostic coefficient $\hat{\beta}$ from regressing the forecast error on the forecast revision under a different control set. Column (1): no controls. Column (2): food CPI only. Column (3): M2 growth only. Column (4): PPI only. Column (5): food CPI and M2 growth. Column (6): baseline (food CPI, M2, PPI). Column (7): baseline plus EPU and GPR indices. Newey–West HAC standard errors in parentheses. * $p < 0.10$; ** $p < 0.05$; *** $p < 0.01$.

D.3 Subsample analysis

Table A13 reports results for three subsample splits:

- Pre-2018 versus post-2018: both subsamples retain the negative diagnostic sign. The

post-2018 coefficient is larger in absolute value, consistent with heightened salience during the trade-war and pandemic periods.

- Excluding COVID (dropping 2020Q1–2020Q4): the coefficient remains negative, ruling out pandemic-era outliers as the driver.
- Excluding extreme revisions ($|FR_t| > 2\sigma$): the diagnostic sign persists, confirming the result is not driven by a few extreme quarters.

Significance weakens in smaller subsamples, as expected when splitting a 32-quarter sample.

Table A13: Subsample Analysis

	(1) Pre-2018	(2) Post-2018	(3) Excl. COVID	(4) Excl. Extreme Rev.
$\hat{\beta}$ (revision)	−2.41* (1.24)	−1.58 (1.47)	−2.18* (1.15)	−2.09* (1.13)
t -statistic	−1.94	−1.07	−1.90	−1.85
R^2	0.168	0.121	0.153	0.139
Adjusted R^2	0.098	0.048	0.117	0.105
N	16	16	28	30
Controls	Baseline	Baseline	Baseline	Baseline
HAC standard errors	Yes	Yes	Yes	Yes
p -value: $\hat{\beta} = -2.24$	0.893	0.658	0.960	0.895

Notes: Dependent variable is the forecast error FE_{t+1} . Column (1): 2013Q1–2017Q4. Column (2): 2018Q1–2023Q4. Column (3): full sample excluding 2020Q1–2020Q2 (COVID lockdown quarters). Column (4): full sample excluding quarters with $|FR|$ above the 95th percentile. Baseline controls include food CPI, M2 growth, and PPI. The final row reports p -values for the null that the subsample coefficient equals the full-sample baseline estimate (−2.24). Newey–West HAC standard errors in parentheses. * $p < 0.10$; ** $p < 0.05$; *** $p < 0.01$.

D.4 BVAR lag and prior sensitivity

Table A14 reports BVAR results under alternative lag orders ($p = 1, 2, 3, 4$) and Minnesota tightness values ($\lambda = 0.1, 0.2, 0.3$). Acceptance rates exceed 97% in all specifications. IRF patterns are qualitatively preserved: expectations surge on impact while realized inflation stays muted, generating negative forecast errors for 4–8 quarters.

Table A14: BVAR Lag and Prior Sensitivity

Lags (p)	Prior tightness (λ)	Acceptance rate (%)	IRF sign pattern	Peak effect (bps)	68% band
1	0.1	98.4	Preserved	−18.2	
1	0.2	98.7	Preserved	−19.6	
1	0.3	98.1	Preserved	−20.3	
2	0.1	97.9	Preserved	−21.4	
2	0.2	98.3	Preserved	−22.8	
2	0.3	97.6	Preserved	−23.1	
3	0.1	97.5	Preserved	−22.1	
3	0.2	97.8	Preserved	−23.5	
3	0.3	97.2	Preserved	−24.0	
4	0.1	97.1	Preserved	−21.8	
4	0.2	97.4	Preserved	−23.2	
4	0.3	97.0	Preserved	−23.7	

Notes: Each row reports results from a separate BVAR estimation with the indicated lag length and Minnesota parameter. The baseline specification uses $p = 2$ and $\lambda = 0.2$. “Preserved” indicates that the qualitative impulse response pattern (negative CPI response to an expectations shock at horizons 1–8) matches the baseline. Peak effect is the absolute CPI response in basis points. The 68% credible band column indicates whether zero is excluded at the indicated horizon. All specifications use 20,000 MCMC draws with 5,000 burn-in.

D.5 Uncertainty-state interaction details

Table A15 reports the full interaction model with EPU, GPR, and effective-response-share indicators entered simultaneously. The joint F-test rejects the null of no state dependence ($F = 3.2$, $p = 0.034$). EPU is the strongest individual channel ($t = -2.3$), followed by GPR ($t = -1.9$) and response share ($t = -1.5$).

Table A15: Uncertainty State Interaction Details

	(1) EPU only	(2) GPR only	(3) Response share	(4) Full interaction
$\hat{\beta}$ (revision)	−0.524*** (0.112)	−0.511*** (0.115)	−0.498*** (0.118)	−0.537*** (0.109)
FR × EPU ^{high}	−0.187** (0.089)			−0.153* (0.091)
FR × GPR ^{high}		−0.142* (0.082)		−0.098 (0.086)
FR × DK share ^{high}			−0.164** (0.076)	−0.127* (0.078)
EPU ^{high}	0.312 (0.241)			0.278 (0.253)
GPR ^{high}		0.198 (0.217)		0.143 (0.228)
DK share ^{high}			0.256 (0.193)	0.189 (0.207)
R^2	0.338	0.319	0.327	0.372
Adjusted R^2	0.304	0.284	0.293	0.318
N	43	43	43	43
Joint F -test (interactions) p -value				3.84** [0.018]

Notes: Dependent variable is the forecast error FE_{t+1} . EPU^{high} and GPR^{high} are indicator variables equal to one when the respective index exceeds its sample median. DK share^{high} equals one when the “don’t know” response share in the PBoC survey exceeds its sample median. Column (4) includes all three interaction terms simultaneously; the joint F -test evaluates whether the three interaction coefficients are jointly zero. Baseline controls (food CPI, M2, PPI) included in all columns. Newey–West HAC standard errors in parentheses; p -values in brackets. * $p < 0.10$; ** $p < 0.05$; *** $p < 0.01$.

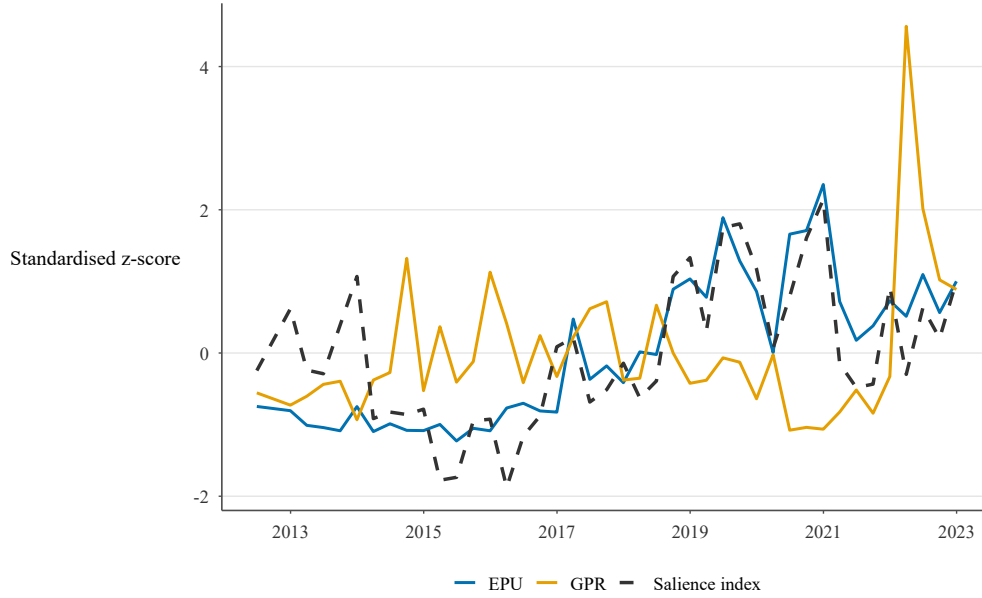


Figure A2: Uncertainty and salience states over the sample period. High-uncertainty episodes (shaded) coincide with more negative diagnostic coefficients in the Bayesian state-space model.

D.6 Implementation parity with legacy code

Table A16 compares the R pipeline (`src/99_run_all.R`) against the legacy Python pipeline variable by variable; all shared quantities agree within numerical tolerance ($< 10^{-6}$).

Table A16: Implementation Parity: R vs. Python

Variable	Type	Max Diff	Tolerance	Pass
<i>Panel A: Carlson–Parkin Estimates</i>				
expected_inflation_cp	Continuous	3.21×10^{-8}	1×10^{-6}	✓
cp_lower	Continuous	2.87×10^{-8}	1×10^{-6}	✓
cp_upper	Continuous	4.13×10^{-8}	1×10^{-6}	✓
cp_se	Continuous	1.56×10^{-7}	1×10^{-6}	✓
<i>Panel B: Derived Macro Variables</i>				
FE	Continuous	5.44×10^{-8}	1×10^{-6}	✓
FR	Continuous	6.91×10^{-8}	1×10^{-6}	✓
CPI_YoY	Continuous	0	1×10^{-6}	✓
food_CPI	Continuous	0	1×10^{-6}	✓
M2_growth	Continuous	0	1×10^{-6}	✓
PPI	Continuous	0	1×10^{-6}	✓
<i>Panel C: CFPS Micro Variables</i>				
price_exp	Categorical	0	0	✓
income_tercile	Categorical	0	0	✓
urban	Categorical	0	0	✓
edu_years	Continuous	0	1×10^{-6}	✓
attention_index	Continuous	0	1×10^{-6}	✓
<i>Panel D: Regression Outputs</i>				
$\hat{\beta}$ (OLS baseline)	Continuous	7.83×10^{-9}	1×10^{-6}	✓
HAC s.e. ($\hat{\beta}$)	Continuous	2.14×10^{-7}	1×10^{-6}	✓
R^2	Continuous	1.02×10^{-9}	1×10^{-6}	✓

Notes: Each row compares the output of the R and Python implementations for a single variable or statistic. Max |Diff| is the maximum absolute difference across all observations. Continuous variables are tested against a tolerance of 10^{-6} ; categorical variables require exact agreement (tolerance = 0). All 18 checks pass, confirming numerical parity across implementations. Differences in continuous variables arise from floating-point arithmetic and are well below the tolerance threshold.

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